

chrna3 Modulates Alcohol Response

Joshua Raine,¹ Caroline Kibat,^{1,2} Tirtha Das Banerjee,³ Antónia Monteiro,³ and  Ajay S. Mathuru^{1,2,4,5,6}

¹Department of Physiology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore 117593, Singapore, ²Yale-NUS College, National University of Singapore, Singapore 138527, Singapore, ³Department of Biological Sciences, National University of Singapore, Singapore 117558, Singapore, ⁴Institute for Digital Medicine (WisDM), Yong Loo Lin School of Medicine, National University of Singapore, Singapore 117456, Singapore, ⁵Institute of Molecular and Cell Biology, A*STAR, Singapore 138673, Singapore, and ⁶Healthy Longevity Translational Research Programme, Yong Loo Lin School of Medicine, National University of Singapore, Singapore 117596, Singapore

Alcohol use disorders (AUDs) are complex phenomena governed by genetics, neurophysiology, environment, and societal structures. New methods to understand the underlying neurogenetics are valuable for designing personalized interventional strategies. Here, we used a two-choice self-administration zebrafish assay to isolate the function of nicotinic acetylcholine receptor subunit alpha3 (*chrna3*) in alcohol response. Juvenile zebrafish, of either sex, prior to complete sex differentiation, were examined in this study. They exhibited a biphasic response when self-administering alcohol that transitioned from attraction to aversion within minutes, suggesting they can regulate exposure to alcohol. This inverted U-shaped self-administration mirrored the effect alcohol has on shoaling behavior. Exposure to low concentration of alcohol reduced anxiety-like behaviors, while sedative effects became prominent at higher concentrations resulting in reduced locomotion and uncoordinated swimming. In contrast, these responses are blunted in *chrna3* mutants. They exhibited prolonged alcohol self-administration and increased gregariousness. Transcriptomic analyses suggest that glutamatergic and GABAergic neurotransmission alongside cholinergic signaling is impacted in the mutant brains. Our results thus suggest that *chrna3* dysfunction has a systemic change with an increase in alcohol tolerance being one effect. These findings also highlight the use of nonrodent alternatives to understand the neurogenetics of development of AUD.

Key words: alcohol; hormesis; nicotine; preclinical model; substance use disorder; zebrafish

Significance Statement

This study sheds light on how a specific gene, *chrna3*, influences the body's response to alcohol. Using a novel self-administration zebrafish assay, we first discovered that zebrafish acute response to alcohol is biphasic. Changes in this *chrna3* function can alter sensitivity and preference to alcohol. This is important because it helps elaborate on the genetic variance among people making them more or less susceptible to alcohol dependence. Our findings also suggest that this gene plays a role in communication pathways within the brain, potentially impacting other related conditions. Ultimately, this work offers a new avenue for building empirically tested knowledge of genetic predisposition and paves the way for future personalized treatment plans.

Introduction

Substance use disorders (SUDs), encompassing nicotine, alcohol, and opioid abuse, represent a significant global health burden, accounting for ~162.5 million disability-adjusted life years lost worldwide (Rehm and Shield, 2019a; Moeller and Terplan, 2020). According to the WHO, harmful alcohol use alone is

estimated to have resulted in 5.3% of deaths across the world in 2022 (Witkiewitz et al., 2019; Rehm and Shield, 2019b; Aslam and Kwo, 2023). Efforts to understand the interplay among social, environmental, and biological factors to effectively counter alcohol use disorders (AUDs) continue to be a focus area, as the response to alcohol in vertebrates itself is a complex

Received Feb. 12, 2025; revised Aug. 2, 2025; accepted Aug. 17, 2025.

Author contributions: J.R. and A.S.M. designed research; J.R., C.K., and T.D.B. performed research; C.K., T.D.B., A.M., and A.S.M. contributed unpublished reagents/analytic tools; J.R. analyzed data; J.R. and A.S.M. wrote the paper.

We thank the zebrafish fish facility (ZFF) staff at the IMCB, A*STAR, for their assistance with fish husbandry. We also thank Li Xinrui for generating customized Python analysis scripts. Additionally, during the preparation of this work, we used Grammarly and Gemini to assist with grammar and consistency of phrasing. After using this tool/service, we reviewed and edited the content as needed and take full responsibility for the content of the publication. This research was supported by the Ministry of Education, Singapore (MOE; through Grant

Number T2EP30220-0020), Yale-NUS College (through Grant Numbers IG16-LR003, IG18-SG103, IG19-BG106, and SUG), Institute for Digital Medicine (WisDM), and Dean's Office, Yong Loo Lin School of Medicine, National University of Singapore, grants to A.S.M.

The authors declare no competing financial interests. The sponsors had no role in the design, execution, interpretation, or writing of the study.

Correspondence should be addressed to Ajay S. Mathuru at ajay.mathuru@nus.edu.sg.

This paper contains supplemental material available at: <https://doi.org/10.1523/JNEUROSCI.0304-25.2025>

<https://doi.org/10.1523/JNEUROSCI.0304-25.2025>

Copyright © 2025 the authors

multifactorial phenomenon. Alcohol interacts with a range of neuromodulatory systems, engages reward processing circuits, and affects multiple neurotransmitter systems, including glutamate and GABA neurotransmission (Chvilicek et al., 2020). Both genetic and neurobiological factors contribute significantly to AUD development (Egervari et al., 2021; Koob and Vendruscolo, 2023). Thus, a comprehensive understanding of these factors, while accounting for individual variability, is crucial for developing effective AUD prevention and interventional strategies (Witkiewitz et al., 2019; Yang et al., 2022; Aslam and Kwo, 2023).

Animal models are indispensable for dissociating genetic contributors to the development of substance dependence in humans (Smith, 2020). While such studies are typically conducted in mammalian models, high operating costs can hinder studies on genetic variability and small, single gene effects (Cox, 2015; de Magalhães, 2015). Zebrafish present a cost-effective alternative, combining high-throughput genetic manipulation with whole-brain live neural activity imaging (Cully, 2019). Additionally, anatomical and functional conservation with mammalian subcortical circuitry is now well established (Kily et al., 2008; Gerlai, 2020).

Zebrafish exhibit behavioral and physiological responses to psychoactive substances that are consistent with mammals, providing construct validity; however, studies have primarily employed indirect measures, like place preference or avoidance assays (Echevarria et al., 2011; Tsang et al., 2019). These are useful but are considered inadequate to model SUD (Smith, 2020). Despite the difficulty in designing contingent, self-administration assays for aquatic animals, they are essential to gain predictive validity (Smith, 2020). Few studies have explored active choice (Schneider et al., 2023) or self-administration (Sterling et al., 2015; Bossé and Peterson, 2017; Nathan et al., 2022). Among these, studies with juveniles (~4–5 weeks), as reported here, offer an advantage over both larvae and adults as behavioral responses become more consistent by this age and a relatively large number of individuals can be examined to detect small effects.

Here, we expanded on the self-administration zebrafish assay (SAZA; Nathan et al., 2022) to study *chrna3*'s role in acute, dynamic alcohol responses. Human genetic studies consistently associate single-nucleotide polymorphisms (SNPs) in the *CHRNA5-A3-B4* cluster, encoding nicotinic acetylcholine receptor (nAChR) subunits $\alpha 5$, $\alpha 3$, and $\beta 4$, with nicotine addiction by GWAS (Thorgeirsson et al., 2008; Amos et al., 2010; Bierut, 2011). Rare variants in *CHRNA3* have been linked to alcohol dependence in humans (Haller et al., 2014; Kamens et al., 2023). While investigations into contributions of *CHRNA5* and *CHRNA4* to SUD have progressed rapidly using rodent models (Fowler et al., 2011; Frahm et al., 2011), such models for *CHRNA3* are fewer due to neonatal lethality of knock-outs (Xu et al., 1999; Elayouby et al., 2021). This leaves the role of *CHRNA3* in alcohol dependence unclear and correlational. Our study fills this gap, validating *chrna3*'s contribution to the acute alcohol response.

Using SAZA for juvenile zebrafish, we report an acute biphasic response to alcohol, transitioning rapidly from attraction to avoidance. This correlates with the proposed role of alcohol as both stimulant and sedative. *chrna3* mutants showed alcohol tolerance, blunting the avoidance response and resisting changes to shoaling behavior. In the context of the Genetically Informed Neurobiology of Addiction (Bogdan et al., 2023), our study provides new evidence that variations in

CHRNA3 may directly influence predisposition to alcohol addiction in humans.

Materials and Methods

Fish husbandry. Zebrafish (*Danio rerio*, ABWT) were housed, bred, and reared as described by Nathan et al. (2022) at the ZebraFish Facility (Institute of Molecular and Cell Biology, A*STAR) in groups of 20–30 in 3 L tanks under standard facility conditions. All experimental protocols involving zebrafish were approved by the Institutional Animal Care and Use Committee (IACUC) of the Biological Resource Center at A*STAR. Approved experimental protocols (IACUC #201529, #231808) were followed.

Generation of *chrna3* mutant zebrafish. The full details of zebrafish gene editing by CRISPR/Cas9 have been described by Hwang et al. (2013). CRISPR targets were determined using the web tool “chop-chop” for the Exon 5 region to disrupt key functional domains. Oligonucleotides containing the identified CRISPR target sequence (GGN₂₀), along with a universal reverse primer encoding the sgRNA scaffold, were used to generate a transcription template, following the method described by Bassett et al. (2013). The sequences of these oligonucleotides are listed in Table 1. CRISPR target sites are underlined, and homology arms are shown in lowercase letters. Zebrafish embryos (nacre background) were injected at the one-cell stage with 1 nl of a mixture containing 1 μ l sgRNA ($\approx$ 4–5 μ g) and 1 μ l Cas9 mRNA ($\approx$ 1 μ g). Fish were genotyped by fin-clipping and sequencing of genomic DNA at 10 weeks postinjection to identify founder fish (F0). The identified founder fish were outcrossed to AB background wild-type (WT) fish to obtain F1 embryos. F1 embryos from each outcrossed family were collected, and some of the embryos were verified by sequencing. Once germline transmission was confirmed by Sanger sequencing, the remaining embryos were grown to adulthood. Homozygous *chrna3*^{C246X/C246X} mutants (AB $\times$ nacre mixed background) from the F3 generation onward were used for all subsequent experiments, alongside WT equivalents, and are henceforth referred to as *chrna3*^{C246X}.

Self-administration for the zebrafish assay (SAZA) apparatus setup. The SAZA assay apparatus was set up as described in Nathan et al. (2022), outlined briefly as follows. The assay tank was constructed of opaque, 3-mm-thick acrylic to be 35 $\times$ 75 $\times$ 30 mm (width $\times$ length $\times$ height). One end of the zone contained a dividing sheet of the same material, 30 mm in length, to split the zone into two halves while still permitting access (Fig. 1A). The zone was placed on an LED lightbox for illumination, providing 5,000 lux at maximum intensity (LightPad LX Series, Artograph). The video footage was captured at 30 fps using an acA2040-90 μ m USB3.0 Basler camera. Solutions were dispensed by gravity from vertically mounted 10 ml syringes through silicon tubing (outer diameter, 1/16"; inner diameter, 1/32") into the corner of their designated zone, mediated via solenoid pinch valves (AutoMate Scientific, SKU: 02-pp-04i; Fig. 1A). System water was consistently supplied to the tank through additional silicon tubing at the opposite end of the tank to the divided zones and was removed at a matching rate ($\sim$ 2 ml per minute) by gravitational siphoning from the end of the dividing sheet near the center of the tank (Fig. 1A). The inflow and outflow were adjusted by gravity feed to maintain the integrity of the two delivery zones throughout the assay period. This created a consistent flow and stopped dispensed solutions from escaping their designated zones while permitting access to the fish being assayed (Nathan et al., 2022). CRITTA (<http://www.critta.org>), a custom LABVIEW software, was used to track fish movements as described (Krishnan et al., 2014). Subject fish behavior was monitored by a closed-loop, video acquisition program developed in-house that allowed for self-administration of a stimulus or a control when subject fish entered a delivery zone from a neutral zone (Fig. 1A; Movie 1). The tracking system was configured so that only the entry of the fish to a designated region of interest would trigger the pinch valve to release and dispense the corresponding solution.

Alcohol concentration measurement. Alcohol concentrations in the SAZA stimulus zone were measured using a DensitoPro (Mettler

Table 1. Primers for single-guide RNA template

Gene	Label	Primer 5'→3'
<i>chrna3</i>	CRISPR1 (F)	aattaatacagactcactataggCTGCTTCTCATCTCTCTCggttttagagctagaataagc
<i>chrna3</i>	CRISPR2 (F)	aattaatacagactcactataggCCGGTTTCAGCAAACAGACggttttagagctagaataagc
<i>chrna3</i>	CRISPR Universal (R)	TTTGCACCGACTCGGTGCCACTTTTCAAGTTGATAACGGACTAGCCTATTTTAACTTGCT ATTTCTAGCTCTAAAC

CRISPR targets are underlined. Homology arms in lower case.

Table 2. Mean time spent by zebrafish in the stimulus chamber during SAZA over an 18 min stimulus-dispensing period ($n = 30$ – 32)

Solution dispensed	Mean time in stimulus chamber(s)
0% alcohol	238
20% alcohol ($\bar{x}$ 0.5%)	220
70% alcohol ($\bar{x}$ 1.5%)	132

Toledo), returning the measured solution to the zone with minimal disturbance. As attempting to measure the concentration of alcohol during a live fish assay would disturb the fish and prevent a natural response, an artificial dispensing scheme of 10 continuous seconds per minute followed immediately by measurement was used as a substitute (Fig. 1C). This was based on the average time fish spent in the stimulus zone under assorted alcohol dispensing solutions over 18 min of exposure (Table 2). After 12 min, the dispensing cycle was halted, but measurements continued to estimate the effect the outflow rate had on concentration in the zone if no stimulus was dispensed for a long period.

The assay was repeated three times per concentration of alcohol (20, 70%).

SAZA. SAZA was performed as described in Nathan et al. (2022), outlined briefly as follows. First, 40 ml of system water was added to the tank, the in- and outflow tubes were inserted into their respective conditions, and the flow of system water started. One zebrafish of 30–35 d postfertilization (dpf), of either sex, prior to complete sex differentiation, was taken directly from its husbandry tank and added gently to each SAZA tank and then given ~5 min to adjust and recover post-transfer until consistent swimming behavior had resumed and the fish had explored all zones of the SAZA tank. To avoid intrinsically preferred swimming direction bias, we pseudorandomly assigned the stimulus and control delivery zones in each trial such that half of the fish in each group received the stimulus (alcohol) in the left and the other half in the right delivery zone (Fig. 1A). The recording and tracking scheme was then started for 24 min, consisting of three consecutive periods; 3 min of pre-exposure, 18 min of self-administration stimulus delivery, and 3 min postexposure. Only during the 18 min stimulus delivery period would entry to a designated zone trigger dispensing of the corresponding solution (Fig. 1A). The valve opening time was set to a minimum of 5 s per entry. The stimuli delivered were 0/20/70% absolute alcohol diluted in the facility system water. Following each assay, the zone was emptied of all liquid, washed, and refilled before adding the next fish. The volume of each solution dispensed was also recorded. To reduce bias, the zone designated to deliver the stimulus solution was randomly assigned to be either the left or right and changed periodically throughout the experiment such that ~50% of the fish were tested in each configuration. Around 30 fish naive to SAZA were assayed per concentration of alcohol. Following each assay, the fish were killed. Juvenile fish, ~5 weeks old, were used for all behavioral experiments as this developmental stage offers the advantages of more complex locomotor behavior and reduced individual variability over younger larvae while possessing comparable structural brain development to adult fish yet still facilitating higher throughput and practicality of assay design.

SAZA data analysis. SAZA data outputs from CRITTA were processed by custom Python scripts. These provided data on time spent in each zone and mean velocity in each zone used in these analyses. The data were separated into three major time groups; pre-exposure, total stimulus delivery period, and postexposure. The stimulus delivery period

was then further subdivided into either start or end (0–9 m/9–18 m) or 3 min segments (0–3 m/3–6 m/6–9 m, etc.) for further analyses.

The preference index (PI) was calculated as the ratio of the difference between the times spent in the stimulus and control zones relative to the total time spent in the two zones. The PI for a given duration (PI_X) was calculated by the following formula: $PI_X = (TinS - TinC) / (TinS + TinC)$, where $TinS$ is the time in stimulus zone(s) and $TinC$ is the time in control zone(s). This gives the value for PI maximum bounds of + or –1 indicating more time spent relatively in the stimulus or control, respectively. For example, a PI of +1 would indicate 100% of relative time spent in the stimulus zone during one time period. Delta PI between two time periods (ΔPI_{X-Y}) was calculated as $\Delta PI_{X-Y} = PI_Y - PI_X$, where PI_X is the earlier period and PI_Y is the later period. This gives the value for ΔPI maximum bounds of + or –2, with the values indicating the direction and magnitude of the change in preference from the first time point to the second. For example, a ΔPI of +2 would indicate a change from 100% relative time spent in the control zone in the first period to 100% relative time spent in the stimulus zone in the second. Sparklines were also generated to display the number of entries to each zone and the duration of each (Fig. 1A).

Shoaling cohesion assay. The shoaling cohesion assay apparatus was set up as described for SAZA, with the following alterations. The tank had no dividing sheet or inflow, outflow, and stimulus-dispensing tubing (Fig. 4A). Additionally, the tracking of fish movements was not performed. To perform the assay, the tank was half-filled with 20 ml of system water, and then four zebrafish of 30–35 dpf, of either sex, prior to complete sex differentiation, were taken directly from their husbandry tank and added gently to each shoaling cohesion tank. The fish were then given ~5 min to adjust and recover post-transfer until consistent swimming behavior had resumed. Following this acclimation period, 20 ml of 2× concentration alcohol solution was added to the tank to achieve the 1× desired concentration for the assay. Recording began immediately and continued for 30 min. Six repeats were performed per alcohol concentration. Following each assay, the fish were killed.

Shoaling cohesion data analysis. Video analysis was performed in imageJ. First, videos were either trimmed to the first 15 min, then frames were extracted at 2 fps to create an image stack (900 per assay), or one frame per 10 s of footage was extracted from the full 30 min duration (180 per assay). The image stacks for each assay were cleaned using median background subtraction and then converted to 8 bit and auto-thresholded (Fig. 4A). From the thresholded images, particles were detected (size, 100–infinity), and the centerpoint of each was determined. Using the centerpoints, Delaunay triangulation was performed on each image to draw triangles covering the space between all four fish. The area of these triangles was then combined to obtain the total area between the fish, as a measure of shoaling cohesion. These area values were grouped into either total time for the full 30 min duration ($n = 1,080$ per treatment) or 1 min intervals for the first 15 min data ($n = 720$ per time point per treatment). Values over 2,000 mm² were identified as products of processing errors and removed (~2.5%).

As a measure of shoal kinesis, the distance traveled by the shoal centerpoint was calculated using the X and Y coordinates for the centerpoint of the triangulation area, as calculated by imageJ, with the following equation: distance = $\sqrt{[(X_2 - X_1)^2 + (Y_2 - Y_1)^2]}$. X_2 and Y_2 were the subsequent coordinates, 0.5 s after X_1 and Y_1 , respectively. This provided the distance the shoal centerpoint had traveled between each selected frame. Values over 10 mm per 0.5 s were discarded as products of processing errors, as the highest velocities recorded by individual fish were ~20 mm/s (Fig. 3B). This amounted to <5% of the data. These values were grouped into 1 min intervals for the first 15 min data ($n = 720$ per time point per treatment).

Quantitative PCR (qPCR). Twelve-week-old adult zebrafish, of either sex, were killed, and the brains were dissected in PBS, pH 7.0. The brain tissue was homogenized using a microtube homogenizer (Thomas Scientific) in a lysis buffer made up of Trizol (Thermo Fisher Scientific #15596026). RNA was extracted using PureLink Micro Kit (Thermo Fisher Scientific #K310250) according to the manufacturer's protocol. The concentration and purity of the extracted RNA were determined by NanoDrop 2000 Spectrophotometer (Thermo Fisher Scientific). Each brain sample yielded ~100 ng/μl RNA with a ratio of absorbance reading 1.9–2.0 at 260/280 nm. The purified RNA was reverse transcribed by SuperScript II First-Strand Synthesis System (Invitrogen #18091050) using 100 ng/μl of purified RNA to obtain ~1,000 ng/μl of first-strand cDNA. To check for genomic DNA contamination, a negative RT (no reverse transcriptase) was set up for each sample.

The cDNA was diluted with nuclease-free water to 100 ng/μl. The qRT-PCR amplification mixtures (20 μl) contained 100 ng of cDNA, 10 μl 2× GoTaq qPCR Master Mix (Promega #TM318), and 300 nM forward and reverse primer. The primers against *chrn* targets used were designed using Primer3 (v.0.4.0) and are detailed in Table 3.

Reactions were performed using a 7500 Real-Time PCR System (Applied Biosystems) in a 96-well plate format. The cycling conditions were as follows: 95°C for 10 min to activate the polymerase, then 40 cycles of 95°C for 15 s, 60°C for 30 s. After completion, a melt curve was run at 65–95°C for 5 s per step. All PCR efficiencies were above 95%. Primer specificity was validated by a single peak on the post-PCR melt curve and single-band after electrophoresis.

The relative gene expression levels were analyzed using the adjusted delta–delta CT method described by Hellemans et al. (2007). All data were analyzed by one-way ANOVA, with genotype as the independent variable and target gene expression, relative to reference genes, as the dependent variable. For mRNA relative expression data, adjusted *p* values were estimated using Dunnett post hoc tests. Effect sizes for all differences in expression were also calculated using Cliff's delta statistics as described below.

Statistical analyses. Data were analyzed and presented following the principles outlined by Ho et al. (2019), to which a *p* value as a measure of significance is improved upon by the supplementary reporting of effect sizes, means, and 95% confidence intervals (Ho et al., 2019). As the assumptions of normality and homoscedasticity recommended for parametric testing could not be consistently met for the SAZA data, most unpaired data were compared with nonparametric two-sided permutation *T* tests, (5,000 reshuffles) with a significance level of 0.05 and unpaired Cliff's delta, while paired or repeated-measure data were compared using by paired permutation tests under the same parameters. Due to the more complex design and higher replicate availability, the shoaling assay longitudinal data were analyzed using a linear mixed-effect model, with the shoal area or centerpoint travel distance as the response variable, alcohol treatment and time as fixed effects, and assay ID as a random effect to account for repeated measures. The residuals indicated the data were slightly skewed but appeared normally distributed and homoscedastic enough to support a parametric test. Analysis was performed in R using the “dabestr” (v2024.12.24) and “lmerTest” (v3.1-3) packages. In the case of multiple comparisons, the Holm–Bonferroni

step down family-wise error correction was applied to the *p* value. All statistical analysis outputs can be found in supplementary tables. These data were presented in the form of Gardener–Altman and Cummings estimation plots. Results are reported as Cliff's delta or Paired mean difference = *X*, 95% CI [lower, upper], *p* = *Y*. For Cliff's delta, effect sizes greater than ±0.4 and a *p* value of <0.01 was considered meaningful and practically relevant in this study. Smaller Cliff's delta effect sizes of ±0.2 to 0.4 with an accompanying *p* value of <0.05 were also considered potentially meaningful but provisional pending experimental reproduction (Cliff, 1993; Goodman et al., 2019).

Hybridisation Chain Reaction, RNA-fluorescent in situ hybridization (HCR RNA-FISH). Hybridisation Chain Reaction RNA-FISH 3.0 (HCR; Choi et al. (2018) probes against *chrna3*, *chata*, and *nrlp1a* were prepared using an in-house Excel-based program and synthesized by IDT (please see Table 4 for details). The in situ hybridization experiments were carried out in a controlled incubation chamber (40°C temperature and >90% humidity). Briefly, 14 dpf zebrafish embryos, of either sex, prior to complete sex differentiation, were killed and fixed in 4% formaldehyde diluted in 1× PBST at room temperature for 60 min with shaking. The larvae were washed with 1× PBST for 3 min × 3 times. Embryos were permeabilized using a detergent solution (1% SDS, 0.5% Tween 20, 150 mM NaCl, 1 mM EDTA, 0.05 M Tris–HCL), pH 7.5, at 37°C for 30 min. They were washed for 3 min for three times each with 1× PBST and then 5× SSC. Larvae were transferred to 30% probe hybridization buffer (30% formamide, 9 mM citric acid, 0.1% Tween 20, 50 μg/ml heparin, 1× Denhardt's solution, 5× SSC) and incubated at 37°C for 30 min. Simultaneously, probes were prepared in a solution containing 30 μl (10 μm) of the probe set against each gene in 1,000 μl of 30% probe hybridization buffer. The larvae were either transferred to the robotic chamber or a 1.5 ml centrifuge tube for automated or manual experiments. Larvae were incubated with the probes at 37°C for 8–12 h. For both methods, on Day 2, larvae were washed five times with 30% probe wash buffer (30% formamide, 9 mM citric acid, 0.1% Tween 20, 50 μg/ml heparin, 5× SSC) at 37°C and then 5× SSC twice. For the secondary reaction, HCR hairpins (Molecular Instruments) were diluted and prepared in an amplification buffer (5× SSC, 0.1% Tween 20, 2.5% dextran sulfate) and incubated overnight (for the manual methodology) or 3 h (for the automated) in the dark. The larvae were then washed with 5× SSC, 4 × 10 min each in the dark. They were mounted with the coverslip on the dorsal side in 70% glycerol–PBS mounting media and imaged on an Olympus FV3000 confocal microscope with the settings consistent between fish. Images were postprocessed first on Olympus viewer FLUOVIEW FV31S-SW software with linear intensity adjustments of 150–4,095 for all genes. Images presented for *nrlp1a*, *chata*, and *chrna3* had no further processing beyond cropping to the ROI and addition of scale bars. The *elavl3* and *chrna3* false color images were processed in ImageJ. The *elavl3* images had additional linear brightness/contrast adjustment (15–110). False color images were converted to 8 bit and had the “fire” LUT applied, followed by linear brightness/contrast adjustment of 0–175, background subtraction with 25 px rolling ball, and 1 px Gaussian blur. Fluorescence quantification (Figs. S1, S2) was performed in ImageJ on the raw images without post-processing. The regions of interest were marked using the boundaries observed in *chata* for the facial motor nucleus (Fmn) or *nrlp1a* for the dorsal habenula, and the mean gray pixel value within the region was measured. Two-week-old (14 dpf) larvae were used for HCR imaging as this age represented the upper limits of transparency without additional optical clearing steps while being closer to behavioral testing age than the traditional imaging of 5–7 dpf larvae. Single-cell analysis shows that 97% of the cell types in a 15 dpf fish are present in 25 dpf fish (Raj et al., 2020), so we deemed it an acceptable compromise for the benefits throughput and visualization over imaging older animals.

RNA-seq. RNA was extracted from 12-week-old adult zebrafish *chrna3*^{C246X} and WT brains, of either sex, in triplicate as described previously. The quality of each sample was verified by gel electrophoresis and bioanalyzer. Samples that passed QC were sent for sequencing by NovogeneAIT. An mRNA cDNA library made up of paired-end sequencing reads of 150 bp in length was generated on the Illumina HiSeq

Table 3. Primers used for qPCR of *chrna3*, *chrna5*, *chrnb4*, and housekeeping genes

Gene accession number	Gene symbol	Primer 5'→3'
β-Actin1 NM_131031.2	<i>actb1</i>	(F) AGATGACACAGATCATGTTCGAGA (R) CCACTAGTAGACACCAAGCG
Elongation factor I-alpha FJ915061.1	<i>ef1a</i>	(F) CTGGAGGCCAGCTCAACAT (R) ATCAAGAAGAGTAGTACCGCTAGCATT
<i>chrna3</i> ENS DARG00000100991.2	<i>chrna3</i>	(F) CCTCCTGTGCTCCGACTGAAC (R) CTCCAGTCATCTCTGACCTC
<i>chrna5</i> ENS DARG00000003420	<i>chrna5</i>	(F) GGATACCAACAGCTCGATG (R) GATGAAAAGCAGAGCAGAGCC
<i>chrnb4</i> ENS DARG00000101677	<i>chrnb4</i>	(F) GAGATTGAGGAGGCTGTGGA (R) CACCACACACGATCACAA

Table 4. Probe sequences for FISH-HCR

Probe name (gene)	Probe sequence
B2_P1 (<i>chrna3</i>)_1	CCTCGTAAATCCTCATCAAACCAAGATTGGTCTCCATAATCTGG
B2_P1 (<i>chrna3</i>)_2	CCTCGTAAATCCTCATCAAAGGATGAATCAACACCTCCAAATC
B2_P1 (<i>chrna3</i>)_3	CCTCGTAAATCCTCATCAAACCTGGAATCTCCACCGCATTGT
B2_P1 (<i>chrna3</i>)_4	CCTCGTAAATCCTCATCAAACCTGAAGTGGCCGGTGGGATCCAG
B2_P1 (<i>chrna3</i>)_5	CCTCGTAAATCCTCATCAAACGAGCCAAATCTCATGGTGACGTT
B2_P1 (<i>chrna3</i>)_6	CCTCGTAAATCCTCATCAAATCCAGAAGTCTTCAGGTGATGG
B2_P1 (<i>chrna3</i>)_7	CCTCGTAAATCCTCATCAAACCTCTACAGCAGTGTATTTAATG
B2_P1 (<i>chrna3</i>)_8	CCTCGTAAATCCTCATCAAATGATCATGTGTGATGGTGAGAAGAG
B2_P2 (<i>chrna3</i>)_1	CTTATAGTCATTCAAATGTGTCTCAAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_2	GCTTCAAATCTTGTGGACGGCACAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_3	CGCAGAAGGGCTTTGGTTTATCGTAAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_4	GTAGTGACGTCGATCTTGACGAGAGAAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_5	GGTCGATCTTGGCCTTGTGCTAGGTAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_6	GGTCGATGATGTCCTTCTCCGCAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_7	CAGTGAATACGTGATGTCCTGTAGAAATCATCCAGTAAACCGCC
B2_P2 (<i>chrna3</i>)_8	TCAGGAAGGAGATGAGAAGACAGGGAATCATCCAGTAAACCGCC
B3_P1 (<i>nrlp1a</i>)_1	GTCCCTGCCTCTATATCTTTGAATCACCCTATGCATTCTGAG
B3_P1 (<i>nrlp1a</i>)_2	GTCCCTGCCTCTATATCTTTCACCTCTCGGCTTCCAGGTCAAAG
B3_P1 (<i>nrlp1a</i>)_3	GTCCCTGCCTCTATATCTTTTCCAAATATTGCCACCAAGCTG
B3_P1 (<i>nrlp1a</i>)_4	GTCCCTGCCTCTATATCTTGTCTGATGTCGACACAACTTGA
B3_P1 (<i>nrlp1a</i>)_5	GTCCCTGCCTCTATATCTTGGTGAAGTCTCGGAACATTCTGGA
B3_P1 (<i>nrlp1a</i>)_6	GTCCCTGCCTCTATATCTTTGAACGTGACGTCAAATATTGGG
B3_P1 (<i>nrlp1a</i>)_7	GTCCCTGCCTCTATATCTTTGCGTGTCTGGCTCCAGTCAAAC
B3_P1 (<i>nrlp1a</i>)_8	GTCCCTGCCTCTATATCTTTGGACCACTCCAGGGAATCCGTC
B3_P2 (<i>nrlp1a</i>)_1	AAAATCTCTGGTGGGCTCGGAGTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_2	GTCTCTACTCCACATAGTCATATTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_3	ACGAGACCCGAGATGGAGCGATTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_4	TAGCGGATGGAGAATCCGGCACCCTTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_5	GGGCGACTTGATGACTCCGCTGCTGTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_6	TTTCTGACATCTAGGAGCAAAGATTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_7	TATCGGCAGAACTCCGGCGGGCTTCCACTCAACTTTAACCCG
B3_P2 (<i>nrlp1a</i>)_8	ATTCTGTCCGAGTATCTGCCGATGTTCCACTCAACTTTAACCCG
B1_P1 (<i>elavl3</i>)_1	GAGGAGGGCAGCAAACGGAATCTCTGCTTCGTTCCGTTGTGCGA
B1_P1 (<i>elavl3</i>)_2	GAGGAGGGCAGCAAACGGAAGTTGGCGAATTTACGGTGATGGGC
B1_P1 (<i>elavl3</i>)_3	GAGGAGGGCAGCAAACGGAAGTGTAGCGGCGAGCGGCTGTCTG
B1_P1 (<i>elavl3</i>)_4	GAGGAGGGCAGCAAACGGAATGGTTATGGGGAGAATCTGGGGA
B1_P1 (<i>elavl3</i>)_5	GAGGAGGGCAGCAAACGGAAGACGAAGTACACGACCGGCTCCA
B1_P1 (<i>elavl3</i>)_6	GAGGAGGGCAGCAAACGGAATGTGACGGCGCCAAAGGCCCGAA
B1_P1 (<i>elavl3</i>)_7	GAGGAGGGCAGCAAACGGAATAGTTGGTCTAGTGACGAAGCCAA
B1_P1 (<i>elavl3</i>)_8	GAGGAGGGCAGCAAACGGAACGAGACCTGCAGCACGGCTGCCCC
B1_P2 (<i>elavl3</i>)_1	TTCTGACCGTTCAGGCCCTTGATGGAAGAAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_2	AGCCTGTCTGTCTTCTGACTGGGTAGAAGAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_3	AGCGCTGGTCTGTGGTGCAGAGGTAGAAGAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_4	ACCCCGGCAAGACTAGTCATGCTGTTAGAAGAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_5	TTCGTACGTTCCGGGACAGGTTGTAGAAGAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_6	TGGTGAAGTACGGATGACCTTGACTAGAAGAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_7	CTGGCGATAGCCATGGCTGCTGTTAGAAGAGTCTTCTTTACG
B1_P2 (<i>elavl3</i>)_8	AGCCTGTGCTGCTGCTGCTGTTAGAAGAGTCTTCTTTACG
B3_P1 (<i>chata</i>)_1	GTCCCTGCCTCTATATCTTTCTGAAGTGTCTCTGGGACGAGG
B3_P1 (<i>chata</i>)_2	GTCCCTGCCTCTATATCTTTGAATCTTTCTGAAGGGTTCCCC
B3_P1 (<i>chata</i>)_3	GTCCCTGCCTCTATATCTTTGTGTTCAAATACATGCTTCAA
B3_P1 (<i>chata</i>)_4	GTCCCTGCCTCTATATCTTTACCTTTGACCTTGAAGTCTGTC
B3_P1 (<i>chata</i>)_5	GTCCCTGCCTCTATATCTTTGAGCAGTCCATCAATTAGAGCTTT
B3_P1 (<i>chata</i>)_6	GTCCCTGCCTCTATATCTTTAAGACCTTGTGTACTGATCCATAC
B3_P1 (<i>chata</i>)_7	GTCCCTGCCTCTATATCTTTTACGCATAACTGTGCTCTCTGA
B3_P1 (<i>chata</i>)_8	GTCCCTGCCTCTATATCTTTGCCGGCGAAAGTTACCATCAGTTC
B3_P2 (<i>chata</i>)_1	AAATTTCTCAACGGGCTGGTCTTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_2	AATTGGCTTTTGTCACTTCTCTTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_3	GGACTGGAGTTGACAGGTAGTGCTATTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_4	AATGAGATTAGCAGCAAATCTGAGGTTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_5	ACTGCCACGAGCATGCTACAGGTTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_6	TTTGTCTGCGAGCTGTAGGAGGTTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_7	CTTACATGCCACTATTATGTTCTTCCACTCAACTTTAACCCG
B3_P2 (<i>chata</i>)_8	GCTGAGGTAAAGGCTTTTTCATTTTCCACTCAACTTTAACCCG

B1 probes were visualized with Alexa Fluorophore 546, B2 probes were visualized with Alexa Fluorophore 647, and B3 probes were visualized with Alexa Fluorophore 488. P1 and P2 probes of the same gene are paired according to numbers 1–8.

instrument. Raw reads were first processed by NovogeneAIT to remove adapter, poly-N sequences, and reads with low quality from raw data. All downstream analyses were based on the clean data with a Phred score of 39 indicating a 99.9% base call accuracy. Further analysis was performed in Partek Flow Explore Spatial Multiomics Data using the Partek Flow software, v11.0. Reads were aligned to GRCh38 genome assembly by STAR aligner, default parameters, and subsequently filtered to a minimum mapping quality score of >30. Gene counts were normalized by the median ratio, followed by differential expression analysis in DeSeq2. Significance

Table 5. Genes included in the custom sets “nicotine dependence” and “alcohol dependence”

Gene set	Gene list
Nicotine dependence	<i>grin1a</i> <i>gabrb4</i> <i>grin2aa</i> <i>grin2bb</i> <i>GRIN2B</i> <i>GABRA2A</i> <i>chrnb2</i> <i>chrnb5b</i> <i>chrna6</i> <i>chrna7</i> <i>FO907089.1</i>
Alcohol dependence	<i>camk4</i> <i>ppp1r1b</i> <i>ddc</i> <i>gnas</i> <i>grin3a</i> <i>npv</i> <i>bdnf</i> <i>grin2b</i> <i>shc3</i> <i>th</i> <i>slc6a3</i> <i>slc18a2</i> <i>kctd10</i> <i>mmab</i> <i>ube3b</i> <i>mapt</i> <i>kansl1</i> <i>tcea2</i> <i>si:dkey-215k6.1</i> <i>pde4bb</i> <i>glrbb</i> <i>rbfox</i> <i>tmem106ba</i> <i>esr1</i> <i>ntrk2b</i> <i>slc6a4a</i> <i>mao</i> <i>comta</i> <i>htr1aa</i> <i>htr1b</i> <i>htr2aa</i> <i>htr1b</i> <i>htr2aa</i> <i>htr2cl1</i> <i>pde4ba</i> <i>glrba</i> <i>tmem106bb</i> <i>ntrk2a</i> <i>slc6a4b</i> <i>comtb</i> <i>htr1ab</i> <i>htr2ab</i> <i>slc6a3</i>
Mammalian anxiety phenotypes	

These genes have been associated with nicotine and alcohol dependence, described by Hu et al. (2018).

thresholds were set as follows: up-/downregulated genes, $p < 0.05$, and fold change, $\geq \pm 2$; provisional up-/downregulated genes, $p < 0.05$, and fold change, $\leq \pm 2$; and nonsignificant genes, $p > 0.05$. Custom gene lists and sets for analysis can be found in Table 5. Gene set enrichment analysis (GSEA) was performed using the GRCz11-GO database, 100 permutations. Significance thresholds were set at $p < 0.05$ and FDR < 0.2 . Given the loss of spatial and structural resolution when using bulk RNA sequencing, this investigation was primarily targeted at identifying brain-wide, compensatory mechanisms that could potentially underlie phenotypic changes or seed predisposition to multisubstance dependence and comorbid disorders; we opted for the “fully developed” brain of the sexually mature adult. These larger adult tissues provided larger quantities of high-quality RNA more consistently than younger animals.

Results

Zebrafish preference for alcohol self-administration in SAZA is biphasic

We refined the SAZA (see Materials and Methods) first reported in Nathan et al. (2022) to improve the temporal resolution for quantifying the dynamic nature of the acute response. The SAZA apparatus created three chemically distinct but physically continuous zones or regions within the assay chamber. When the subject fish entered a delivery zone (stimulus or control) from the neutral zone, the corresponding substance was automatically dispensed during self-administration (Fig. 1A; Movie 1). Sparkline plots in pilot experiments showed that fish administered 20 or 70% alcohol (at source) differentially, by changing the frequency of entry, and duration of time spent in the stimulus area (Fig. 1B; Movie 1). This suggested that fish could self-regulate alcohol administration, and their behavior in the closed-loop setup was dependent on stimulus concentration.

Next, we estimated the stimulus concentration that fish experienced during the assay. Alcohol was dispensed mimicking the behavior of fish in the pilot experiment (Fig. 1B). The maximum concentration plateaued at ~ 40 -fold lower than the source alcohol concentration, which was at least fourfold lower than previous studies (Nathan et al., 2022). Therefore, on an average, self-administering source alcohol at 20 and 70% resulted in the fish experiencing alcohol at ~ 0.5 and $\sim 1.5\%$, respectively, in the assay (Fig. 1C; for 20%, $\bar{x} = 0.52\%$; SD, 0.13%; 70%, $\bar{x} = 1.52\%$; SD, 0.44%). We refer to these two conditions as $\bar{x}0.5\%$ condition and $\bar{x}1.5\%$, respectively, in the rest of this study. The effective concentration reduced by half within a minute of stopping self-administration (Fig. 1C). Thus, both the pre- and postadministration periods served as controls to compare the behavior of animals in the naive pre- and postexposure periods. This range of alcohol exposure is comparable with the concentration of alcohol used in previous zebrafish studies (Kurta and Palestis, 2010; Echevarria et al., 2011; Mathur et al., 2011; Tsang et al., 2019).

The dynamic change in alcohol concentration over time during SAZA is a key advantage for untangling the small effects single genes may have on complex behaviors like alcohol preference. Common variants in *CHRNA3*, for example, are risk factors for developing nicotine dependence, while several rare variants are proposed to influence cocaine and alcohol dependence (Haller et al., 2014; Kamens et al., 2023). *Chrna3* is proposed to mediate alcohol stimulant effects in rodents (Kamens et al., 2009). However, this gene's role in development of alcohol dependence hasn't been tested directly. We hypothesized that SAZA will be effective in examining such a role, if present, in nAChR alpha3 (*chrna3*) mutant generated recently (Kibat, 2020).

The *chrna3* mutant harbors an insertion that causes a frameshift and introduces a premature stop codon within

transmembrane Domain 1 of the gene, resulting in a truncated protein at amino acid position 246 (*chrna3*^{C246X}; Fig. 1D). The mutant protein is predicted to have a large truncation of the transmembrane and the intracellular domains (Fig. 1E) and is thus expected to be damaging to protein function. Quantitative RT-PCR of adult whole brains showed that the mutation resulted in a significant reduction in the expression of *chrna3* transcripts in the mutants (Fig. 1F; WT vs *chrna3*^{C246X}; Cliff's $\delta = -0.848$; 95% CI $[-0.567, -0.962]$; $p \leq 0.001$). In rodents, *Chrna3* expression is restricted. It is expressed in a few neurons in the hippocampus, medial habenula (MHb), interpeduncular nucleus (IPN), ventral tegmental area (VTA), cerebellum, locus ceruleus, and spinal cord (Wada et al., 1989; Gotti et al., 2006). We investigated the expression of *chrna3* in 14 dpf zebrafish larvae using an in-house modified Hybridisation Chain Reaction RNA fluorescence in situ hybridization (HCR) method (Banerjee et al., 2024). As cholinergic function in the hindbrain motor neurons and expression in the facial motor nucleus (Fmn) have also been previously reported (Rima et al., 2020), we quantified *chrna3* expression in the mutants. While expression of *chrna3* was high in the WTs, it was starkly and consistently reduced in the mutants (Fig. 1G; Fig. S1). We also examined expression of *chrna3* in brain regions such as the MHb (dorsal habenula in zebrafish), reported to have coexpression of *CHRNA5-A3-B4* in mammals (Xu et al., 2006; Fowler et al., 2011). Only a few cells in the left dorsal habenula expressed *chrna3* at a low level. The expression levels in the mutant and WT were similarly low (Fig. 1H; Fig. S2).

Having established a robust system for quantifying self-administration and generated *chrna3*^{C246X} mutants, we examined the behavior of juvenile fish in the SAZA at an age when they show consistency across the population (~ 5 weeks old). We first examined the response of WT animals. The pilot (Fig. 1B) suggested the preference changed dynamically over the course of the assay. To capture this, we collapsed the preference behavior of animals over 3 min windows that balanced temporal granularity in relation to the changing alcohol concentration with quantifiable observation periods. The preference was unchanged over time when alcohol was absent (0%; Fig. S3). The same analysis for the $\bar{x}0.5\%$ alcohol condition revealed that fish initially spent significantly more time in the stimulus zone compared with the pre-exposure period (Fig. 2A; 3–6 m; paired mean difference, 7.62; 95% CI $[13.9, 0.331]$; $p_{\text{adj}} = 0.028$). However, after ~ 10 min, fish reversed their preference and spent significantly less time in the stimulus zone (Fig. 2A; 12–15 m; paired mean difference, -6.47 ; 95% CI $[-2.2, -10.1]$; $p_{\text{adj}} = 0.031$). This avoidance continued for the remainder of the assay, even after the self-administration period ended. However, the early attractive portion of this biphasic response was absent in the $\bar{x}1.5\%$ condition (Fig. 2B; 3–6 m; paired mean difference, -1.1 ; 95% CI $[4.26, -6.27]$; $p_{\text{adj}} = 1$). Fish spent less time in the stimulus zone from 6 min onward (Fig. 2B; 6–9 m; paired mean difference, -8.57 ; 95% CI $[-3.22, -13.00]$; $p = 0.012$). Mapping the behavioral changes to the alcohol concentration expected in the stimulus zone revealed that juvenile zebrafish preferred alcohol when the concentration was under 0.5%, while they avoided $\sim 0.7\%$ or higher alcohol (Fig. 2C). Thus, WT zebrafish self-regulated alcohol exposure within a narrow range.

chrna3 mutant fish show higher tolerance to alcohol

We next examined the behavior of *chrna3*^{C246X} mutants in the SAZA. Mutants, similar to WT counterparts, showed no

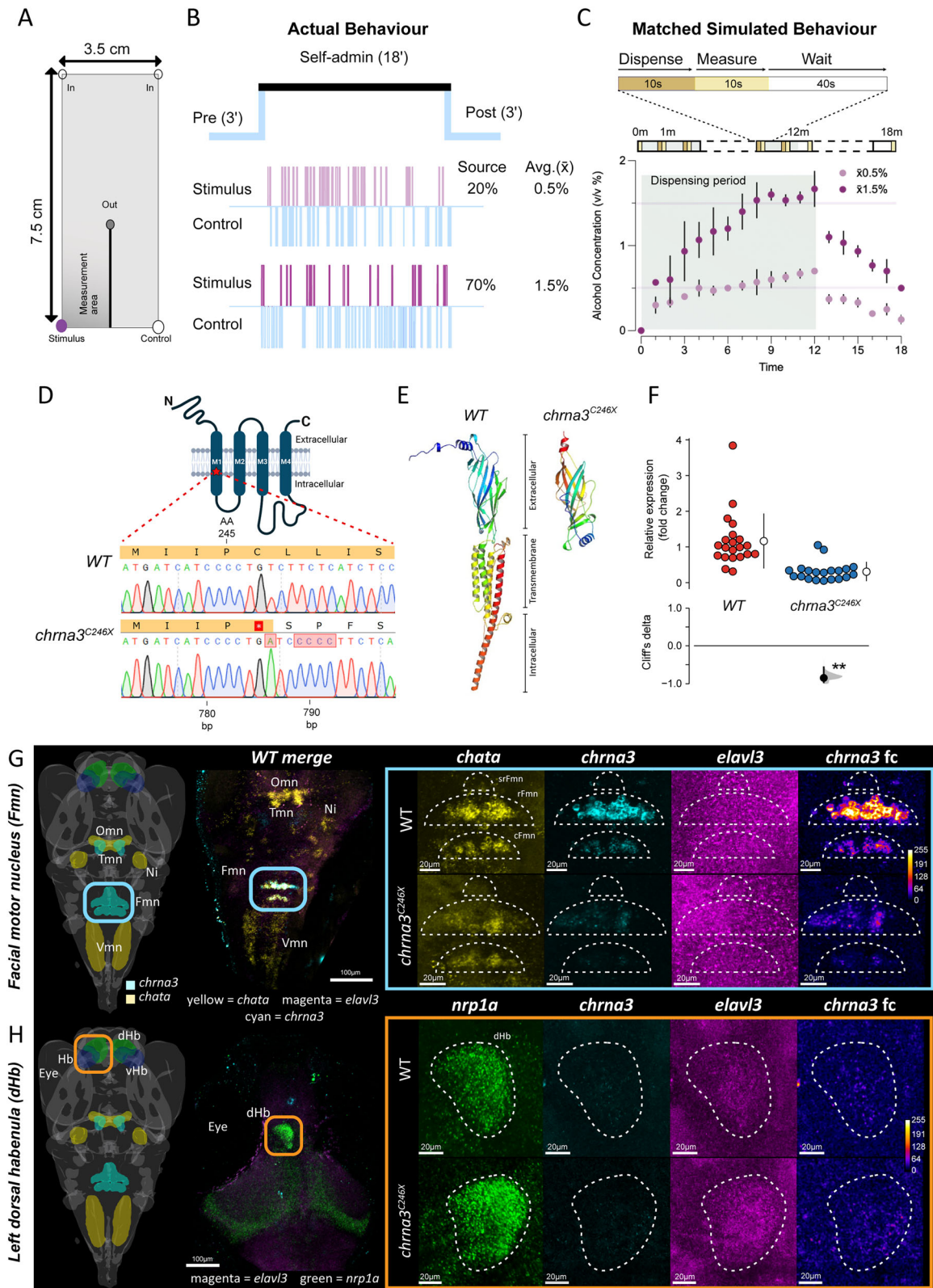
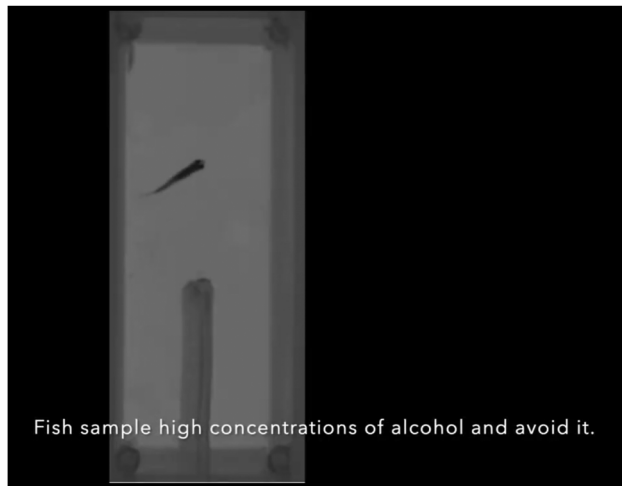


Figure 1. SAZA chamber functional design and characterization of the *chrna3*^{C246X} mutant zebrafish. **A**, Schematic indicating locations of inflow of system water (In, top) and outflow (Out, center). Control and stimulus delivery inlets (stimulus shown in purple) are activated when a fish enters the corresponding zone. Continuous inflow and outflow constrain the stimulus/control delivered to their respective zones (Nathan et al., 2022). "Measurement area" denotes where alcohol concentration was measured in panel **C**. **B**, Schematic shows the experimental design with 3 min pre- and postexposure periods before and after an 18 min self-administration period. Representative sparkline plots show the entries and duration (thickness) of entry of the subject fish either into the stimulus or the control zones when 20 or 70% alcohol is used as stimuli. **C**, Alcohol concentrations measured when 20 or 70% alcohol were dispensed for the duration mimicking subject fish behavior ($\pm$ standard error). The average concentration (solid line) fish experience is 0.5 and 1.5% alcohol, respectively. See Table 2 for additional details. **D–H**, Characterization of *chrna3*^{C246X} mutant fish. **D**, *chrna3*^{C246X} mutants have a premature stop codon in the transmembrane Region 1, amino acid 246 (M1 - red star), due to the indel of 5 bp at position 786 bp. **E**, Predicted protein structure of *chrna3* in WT and *chrna3*^{C246X} mutants, generated by iTasser (Zhou et al., 2022), color coded by domain. **F**, Fold change of *chrna3* in 12-week-old WT ($n = 21$)



Movie 1. A representative segment of the zebrafish self-administration assay (SAZA) with 1.5% alcohol shown in Figure 1B. [View online]

preference in the absence of alcohol (0%; Fig. S3A,B). In the 0.5% condition, *chrna3*^{C246X} fish initially behaved like their WT spending significantly more time in the stimulus zone (Fig. 2D; 3–6 m; paired mean difference, 13.9; 95% CI [20.3, 8.18]; $p_{\text{adj}} = 0.005$). However, unlike the WT, this attraction persisted throughout the assay period (Fig. 2D; 9–12 m; paired mean difference, 5.96; 95% CI [10.1, 1.94]; $p_{\text{adj}} = 0.044$). No biphasic switch to avoidance was observed. Surprisingly, mutants did not show a strong avoidance response during the assay even in the 1.5% condition (Fig. 2E; 12–15 m; paired mean difference, -7.15; 95% CI [-1.13, -12.4]; $p_{\text{adj}} = 0.134$).

The total time spent in the control delivery zone varied among conditions and genotype, but did not show a consistent trend (Fig. S3C–H). The frequency of entry into the stimulus zone, however, was correlated with the time spent in the stimulus zone. This was similar across both genotypes in the 0.5% condition and was indicative of the preference behavior seen (Fig. S4C,D). Interestingly, the frequency of entry into the stimulus zone in the 1.5% condition showed a brief increase initially across the two genotypes (0–3 min; Fig. S4E,F), before it plateaued off to prestimulus frequency rapidly, suggesting that a reduction in total time spent in the stimulus zone observed in Figure 2, B and E, was driven by shorter visits rather than reduced frequency of visits.

Despite the early attraction (Fig. 2A,D), in general alcohol appeared to be aversive to WT zebrafish albeit less so for the mutants. This could also be inferred from the volumes delivered in alcohol conditions compared with the volumes delivered in the 0% alcohol condition (Fig. 3A,B). WT fish self-administered significantly smaller volumes in the 0.5% condition (Fig. 3A; Cliff's $\delta = -0.554$; 95% CI [-0.266, -0.759]; $p_{\text{adj}} \leq 0.001$) and in the 1.5% condition (Fig. 3A; Cliff's $\delta = -0.728$; 95% CI [-0.441, -0.895]; $p_{\text{adj}} \leq 0.001$). This supports previous

observations that alcohol can be aversive to zebrafish at certain concentrations (Nathan et al., 2022). The *chrna3*^{C246X} mutant fish, however, reduced volume of administration only in the 1.5% condition (Fig. 3B; 0% vs 1.5%; Cliff's $\delta = -0.583$; 95% CI [-0.304, -0.791]; $p_{\text{adj}} \leq 0.001$), and the two genotypes differed only during the 0.5% condition (Fig. 3C; 0.5%; Cliff's $\delta = -0.328$; 95% CI [0.584, 0.009]; $p_{\text{adj}} = 0.027$).

To quantify the magnitude of the difference between genotypes, we examined a PI (see Materials and Methods). PI was calculated for each subject as the relative preference for the stimulus compared with control. Positive PI values (0 to +1) therefore indicated a preference for stimulus, while negative values (0 to -1) indicated an aversion. We also quantified a delta or the change in the PI itself in subsequent phases: pre-exposure phase, the early phase (0–9 min), and the late phase (9–18 min). The duration of the phases coincides with the time required (~10 min) for equilibration of blood alcohol concentration (BAC) in larvae immersed in alcohol (Lockwood et al., 2004). We used the “delta–delta” or the difference of this measure between the two genotypes to quantify what aspect *chrna3* contributes to most (Fig. 3).

As anticipated, the PI range was broad, and population mean was centered around zero in the 0% alcohol condition (Fig. 3D). No significant change was observed between pre- and early phases for either genotypes. Consequently, the delta–delta between genotypes was also insignificant (Fig. 3D). However, both WT and mutants showed a positive preference shift in the early phase in the 0.5% condition (Fig. 3E; WT; paired mean difference, 0.181; 95% CI [0.321, 0.017]; $p_{\text{adj}} = 0.036$; Fig. 3E; *chrna3*^{C246X}; paired mean difference, 0.34; 95% CI [0.488, 0.232]; $p_{\text{adj}} \leq 0.001$). The delta–delta between genotypes changed only marginally (Fig. 3E). The genotypic, however, was significantly different in the 1.5% SAZA (Fig. 3F; delta–delta, 0.305; 95% CI [0.478, 0.114]; $p_{\text{adj}} = 0.009$). In the second half of the SAZA self-administration period, once again the 0% condition had no directional change between the phases or the genotypes (Fig. 3G). Although both genotypes showed similar trend in the 0.5% condition (Fig. 3H), it was notable that the individual variance in the PI was small for mutants, while it was considerably broader for WT, distributed across preference and aversion. Finally, in the 1.5% condition, the PI shifted toward aversion or negative values substantially for WT (Fig. 3I; WT; paired mean difference, -0.3; 95% CI [-0.144, -0.462]; $p_{\text{adj}} = 0.001$). On the other hand, the PI in the mutants showed no directionality (Fig. 3I). In addition to this increased alcohol tolerance in the 1.5% condition, the mutants also exhibited heightened alcohol-induced locomotor stimulation over the full stimulus-dispensing period (0.5% condition; Fig. S5A–F) that impacted other behavioral characteristics of the mutants (Fig. S6). Altogether, this suggested that manipulation of *chrna3* function exerted larger effects on the alcohol tolerance at higher alcohol concentrations than impact the attractive response at lower concentrations.

Shoaling behavior shows biphasic effects of alcohol exposure

We next explored the potential reasons for this rapid and biphasic response to alcohol self-administration. Alcohol is well

versus *chrna3*^{C246X} ($n = 20$) by qRT-PCR. **G, H**, *chrna3* expression in 14-d-old zebrafish larvae visualized by Hybridisation Chain Reaction RNA-FISH (HCR) in the **G**. Predicted Fmn marked by *chata* expression and **H** dorsal habenula (dHb) marked by *nrlp1a* expression, in WT and *chrna3*^{C246X}, respectively. Images are representative; see Figures S1 and S2 for replicates (WT/*chrna3*^{C246X} Fmn $n = 3/4$; dHb $n = 4/3$). Schematic images with regional markers (v2.0, MECE, 2024) were generated in mapZebBrain Atlas (mapzebrain.org, May 2025; Kunst et al., 2019). Hb, habenula; vHb, ventral habenula; Omn, oculomotor nucleus; Trm, trochlear motor nucleus; Ni, nucleus isthmi; Vmn, vagus motor nucleus; srFmn, suprarostrol Fmn; rFmn, rostral Fmn; cFmn, caudal Fmn. Genes presented left to right: all genes merged, **G**, *chata*, or **H**, *nrlp1a*, *chrna3*, *elavl3*, and *chrna3* with false color postprocessing. False color is adjusted (fc), and *elavl3* images have been brightened for enhanced visibility.

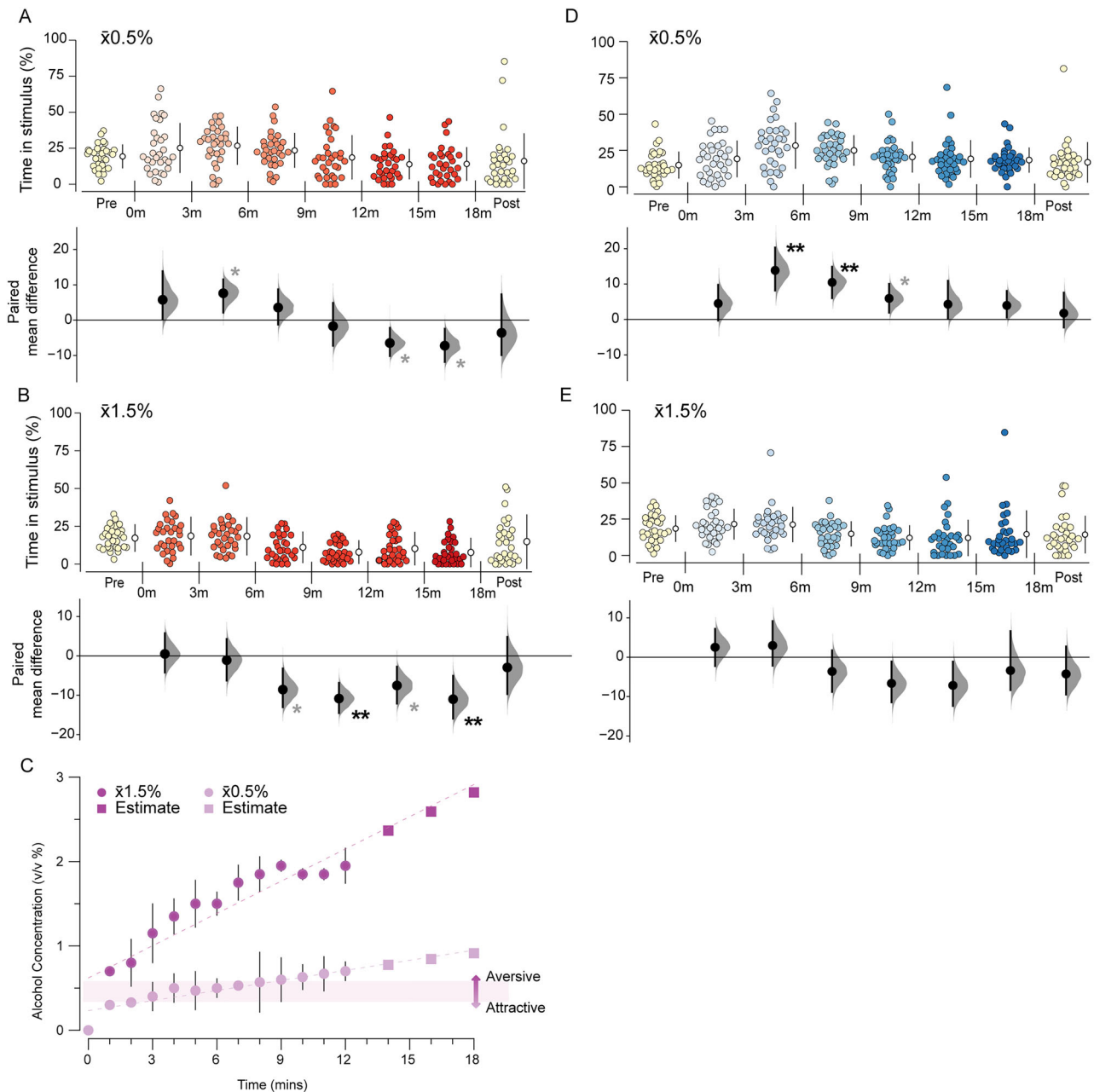


Figure 2. The preference for alcohol is biphasic. Gardner–Altman and Cumming estimation plots show alcohol response in 3 min intervals. No stimulus was dispensed in the “Pre” and “Post” periods. Time spent in the stimulus zone during $\bar{x}$ 0.5% SAZA condition for **A**, WT ($n = 31$), and **D**, *chrna3*^{C246X} mutants ($n = 32$). Time spent in the stimulus zone in the $\bar{x}$ 1.5% SAZA condition for **B**, WT ($n = 30$), and **E**, *chrna3*^{C246X} mutants ($n = 30$). **C**, Ethanol concentrations in the stimulus zone in the $\bar{x}$ 0.5% and $\bar{x}$ 1.5% alcohol conditions and estimated for extended periods. The concentration at which attraction switches to aversion is highlighted. Asterisks indicate a significant paired mean difference between that time portion and the prestimulus period: * $p_{\text{adj}} < 0.05$ (provisional difference), ** $p_{\text{adj}} < 0.01$ (meaningful difference). See Table S1 for the precise effect sizes and p values, corrected for multiple comparisons.

documented to act as an anxiolytic, reducing a variety of anxiety-like behaviors in fish (Gebauer et al., 2011; Araujo-Silva et al., 2020), including reduced shoal cohesion (Miller et al., 2013). However, it is also known to function as a sedative, causing reduced swimming velocity and shoal coordination, and increased immobility (Echevarria et al., 2011; Tsang et al., 2019). We hypothesized that the biphasic response in SAZA is correlated to the concentrations at which these two contrasting effects of alcohol come into play. Hence, the impact of both the anxiolytic and sedative effects of alcohol can be measured by examining shoal cohesion and kinesis in tandem (Fig. 4A).

Taking into account the concentrations associated with attraction and aversion (Fig. 2C), we examined the shoaling behavior of four fish over 30 min at two fixed concentrations of 0.5 or 1% alcohol and compared their behavior when alcohol was absent (0%; Fig. 4A; Movie 2). The total area between WT shoal members increased at both concentrations, with the 1% alcohol having a proportionally larger effect on the shoal area (Fig. 4B; 0 vs 0.5%; Cliff's $\delta = 0.409$; 95% CI [0.453, 0.364]; $p_{\text{adj}} \leq 0.001$; 0 vs 1%; Cliff's $\delta = 0.518$; 95% CI [0.559, 0.475], $p_{\text{adj}} \leq 0.001$). The same phenomenon was also observable in the mutants (Fig. 4C; 0 vs 0.5%; cliff's $\delta = 0.365$; 95% CI [0.412,

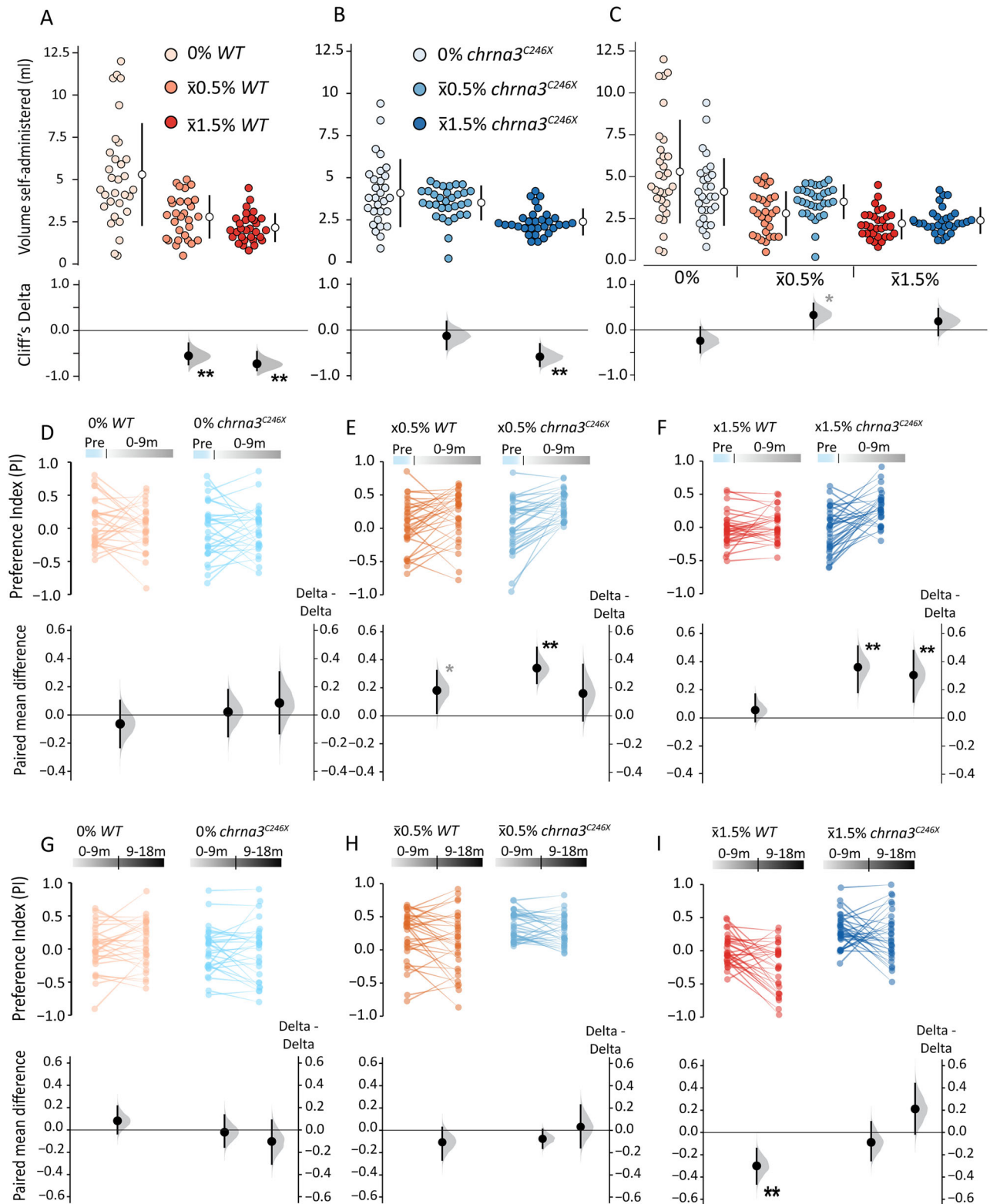


Figure 3. *chrna3* mutation dynamically alters relative preference for alcohol. **A–C**, Gardner–Altman and Cumming estimation plots comparing the volume of alcohol dispensed over the entire self-administration period during SA for **A**, WT (0/0.5/1.5%, $n = 30/31/30$); **B**, *chrna3*^{C246X} (0/0.5/1.5%, $n = 29/32/30$); and **C**, intergroup comparisons. Asterisks indicate the following significant differences from 0% alcohol treatments (**A**, **B**) or the WT (**C**), $p_{\text{adj}} < 0.05$, and the effect size reported by Cliff's delta between ± 0.2 and ± 0.4 (provisional difference), $p_{\text{adj}} < 0.01$ and effect size bigger than ± 0.4 (meaningful difference). **D–I**, Paired mean difference of PI between **D** and **F**, Pre and 0–9 m or **G–I**, 0–9 and 9–18 m time periods within genotypes and delta–delta comparisons between genotypes, during **D** and **G**, 0%; **E** and **H**, 0.5%; and **F** and **I**, 1.5% SA. Asterisks indicate a significant paired mean difference, or delta–delta, between time periods: $p < 0.05$ (provisional difference); $p < 0.01$ (meaningful difference). See Table S2 for the exact effect size and p values.

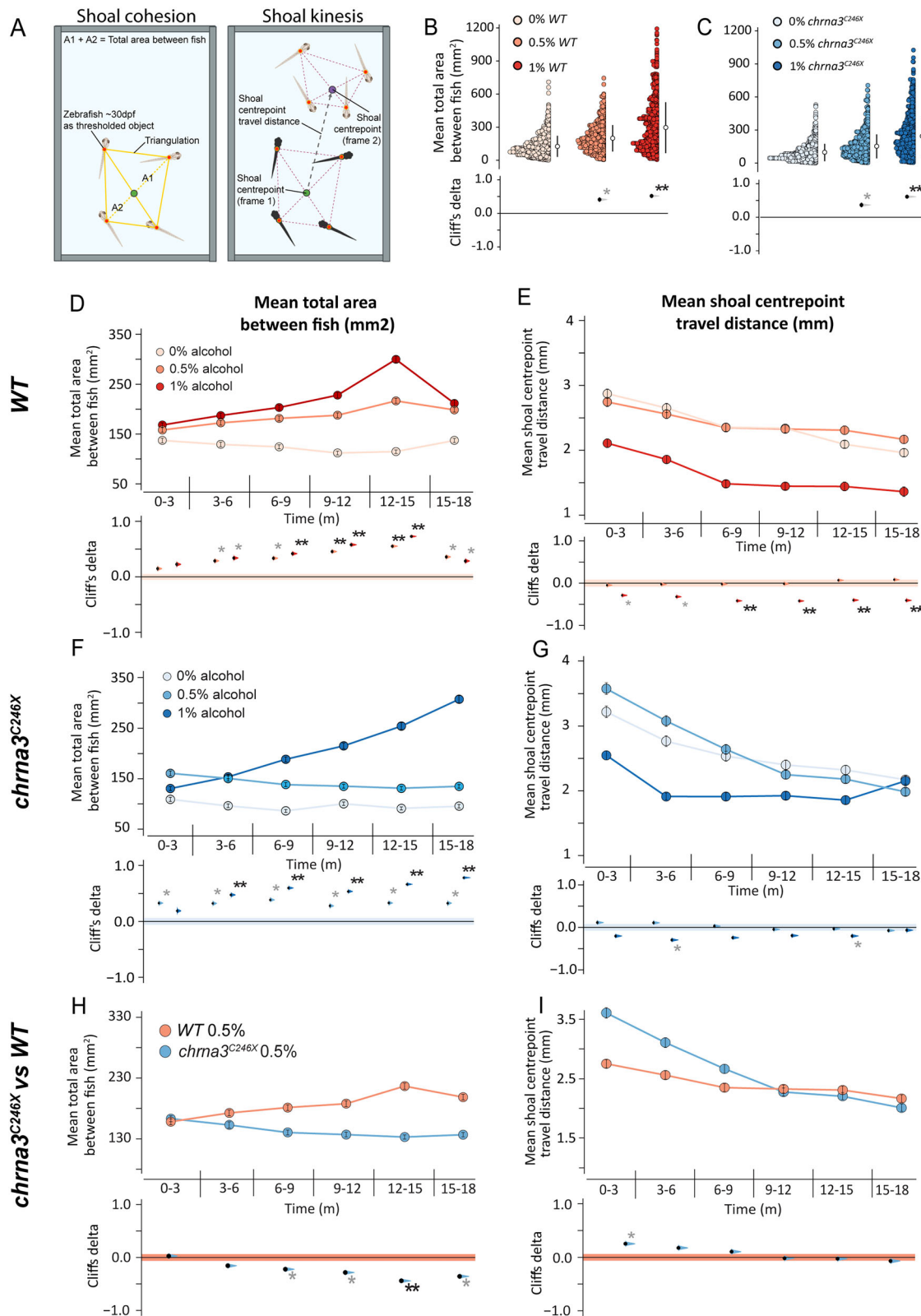


Figure 4. *chrna3* mutants experience reduced psychoactive effects from alcohol exposure. **A**, A schematic outline of the shoaling assay metric calculation and processing steps for cohesion (shoal area) and kinesis (shoal movement). **B** and **C**, Gardner–Altman, or **D**, **F**, and **H**, mean ($\pm 95\%$ CI) shoal cohesion, and **E**, **G**, and **I**, kinesis, with Cumming estimation plots for the shoaling cohesion assay showing **B**, **D**, and **E**, WT; **C**, **F**, and **G**, *chrna3*^{C246X}, and **H** and **I**, intergenotype comparisons. Metrics recorded per frame at 2 fps ($n = 6$ assays per genotype, per treatment). Asterisks indicate a significant difference in the estimated marginal mean from **B** to **G**, 0% alcohol, or **H** and **I**, WT within that time period, * $p_{\text{adj}} < 0.05$, and the effect size reported by Cliff's delta between ± 0.2 and ± 0.4 (provisional difference), ** $p_{\text{adj}} < 0.01$, and the effect size greater than ± 0.4 (meaningful difference). See Table S3 for the exact effect size and p values.



Movie 2. A representative segment of the shoaling assay with 1% alcohol shown in Figure 4D. [View online]

0.320]; $p_{\text{adj}} \leq 0.001$; 1%; Cliff's $\delta = 0.611$; 95% CI [0.649, 0.573]; $p_{\text{adj}} \leq 0.001$).

Differences between the two genotypes emerged when their behaviors were examined at a higher temporal resolution. Loss of cohesion in WT fish was observable within 5 min of exposure to either concentration (Fig. 4D; 0 vs 0.5% 3–6 m; Cliff's $\delta = 0.287$; 95% CI [0.320, 0.252]; $p_{\text{adj}} = 0.032$; Fig. 4D 0 vs 1% 3–6 m; Cliff's $\delta = 0.338$; 95% CI [0.370, 0.305]; $p_{\text{adj}} = 0.004$). The magnitude of the difference increased over time (Fig. 4D; 0 vs 0.5% 12–15 m; Cliff's $\delta = 0.552$; 95% CI [0.579, 0.522]; $p_{\text{adj}} \leq 0.001$; 0 vs 1% 12–15 m; Cliff's $\delta = 0.728$; 95% CI [0.749, 0.707]; $p_{\text{adj}} \leq 0.001$). Shoal kinesis changed negligibly over time with 0.5% alcohol treatment, while it was reduced almost immediately with 1% alcohol treatment. The effect became larger over time (Fig. 4E; 0 vs 1%; 6–9 m; Cliff's $\delta = 0.419$; 95% CI [−0.387, −0.449]; $p_{\text{adj}} = 0.001$). Thus, immersion in 0.5% alcohol reduced anxiety-like behaviors, while sedative-like effects were more evident when immersed in 1% alcohol (Fig. 4D,E).

The *chrna3*^{C246X} mutants seemed to experience the effects of alcohol exposure differently. Behavior of unexposed (0% alcohol) was similar between the two genotypes with the *chrna3*^{C246X} exhibiting only slight differences in shoal cohesion (Fig. S7A) and shoal kinesis (Fig. S7B). However, unlike the WT shoals, exposure to 0.5% alcohol reduced shoal cohesion only marginally (Fig. 4F), and it did not increase over time as it did in the WT's (Fig. 4D). This difference is clearer when genotypes are compared directly. Mutant fish retained significantly tighter shoaling cohesion compared with the WT (Fig. 4H; 12–15 m; Cliff's $\delta = -0.44$; 95% CI [−0.409, −0.47]; $p_{\text{adj}} \leq 0.001$). Meanwhile, shoal kinesis was initially heightened in *chrna3*^{C246X} fish compared with the WT's (Fig. 4I; 0–3 m; Cliff's $\delta = 0.253$; 95% CI [0.288, 0.219]; $p_{\text{adj}} = 0.004$), which paralleled the *chrna3*^{C246X} alcohol-induced motility stimulation in the SAZA (Fig. S5A–F). The alcohol-induced kinesis difference between genotypes gradually declined as the assay progressed (Fig. 4I). This divergence in behaviors between genotypes suggests that the anxiolysis from the alcohol exposure was much lower in *chrna3*^{C246X} than that experienced by their WT counterparts.

The behavior differences between the mutant and WT shoals were minimal in 1% alcohol for the majority of the assay duration (Fig. S7C,D). However, in the latter part of the assay, mutants seemed to resist the sedative effects of alcohol, retaining higher shoal kinesis than the WT (Fig. S7D; 15–18 m; Cliff's $\delta = 0.372$; 95% CI [0.405, 0.34]; $p_{\text{adj}} = 0.008$). The effects of alcohol on shoal cohesion and kinesis in the first 5 min of exposure were similar between genotypes (Fig. 4, Fig. S7). Overall, these results from the shoaling assay suggest that *chrna3*^{C246X} had reduced effect of alcohol exposure on anxiolysis as well as sedative effects seen in WT fish.

Whole-brain transcriptomic analysis shows major neurotransmitter signaling pathways are affected in mutants

To gain an understanding of these behavioral differences, we performed bulk RNA sequencing of WT and mutant whole brains. Differential gene expression analysis suggested that the two genotypes had large differences, with PC1 accounting for ~44% of the variation in a PCA (Fig. 5A). GSEA revealed notable enrichment in transcriptional changes in mutants (Fig. 5B). The significantly enriched biological functions impacted in the mutant included neuronal development, ion channel function, and secondary messenger-associated processes (Fig. 5B). Additionally, genes associated with feeding behavior were also enriched. In total, the expression of ~1,600 genes was significantly changed (two-fold or more), and a further ~4,600 genes were identified with provisional alterations (less than twofold; Fig. 5C). Of these, there was an approximately even split between up- and downregulated gene numbers (Fig. 5C). Together, this perhaps indicates a wide cascade of compensatory changes induced by the reduced expression and function of *chrna3*.

Major neurotransmitter signaling including glutamate and GABA receptors in addition to nAChRs, which are activated by exposure to psychoactive substances like alcohol, also showed modest changes in expression patterns (Fig. 5D–F). Both glutamate and GABA receptors appeared to show a small but consistent upregulation among many receptor subunits (Fig. 5D,E). The expression of most nAChR genes however was unchanged, though a few were positively or negatively regulated with no consistent trend (Fig. 5F). Closer inspection of the nAChR subunit gene counts revealed that expression was very low for some subunits, as was highlighted for *chrna3* in the HCR (Figs. 1, 5G). For genes with low expression counts, it is a challenge to determine significant changes through RNA-seq analysis. As *CHRNA3* expression is coregulated with *CHRNA5* and *CHRNA4* in humans, both of which have also been associated with SUDs (Wen et al., 2016), we examined whether the expression of these genes had also been altered in the mutant. Both *chrna5* and *chrna4* displayed significant downregulation by qPCR, with the effect size being much larger for the latter (Fig. 5H).

We also examined gene sets curated from meta-analyses of genes consistently associated with alcohol and nicotine dependence in humans (Hu et al., 2018) and with mammalian anxiety phenotypes (Howe et al., 2016; Koskinen and Hovatta, 2023) to address whether *chrna3* dysfunction also alters expression of these genes (Fig. 5I–K). The alcohol dependence-associated genes exhibited few changes (Fig. 5I); however, ~50% of the nicotine-dependent genes were provisionally upregulated in the mutant (Fig. 5J). Finally, genes associated with anxiety phenotypes also showed a number of expression changes, albeit some in positive and some in negative direction of expression (Fig. 5K). In all cases, this analysis revealed that though the expression of a large number of these genes changed, they were modest.

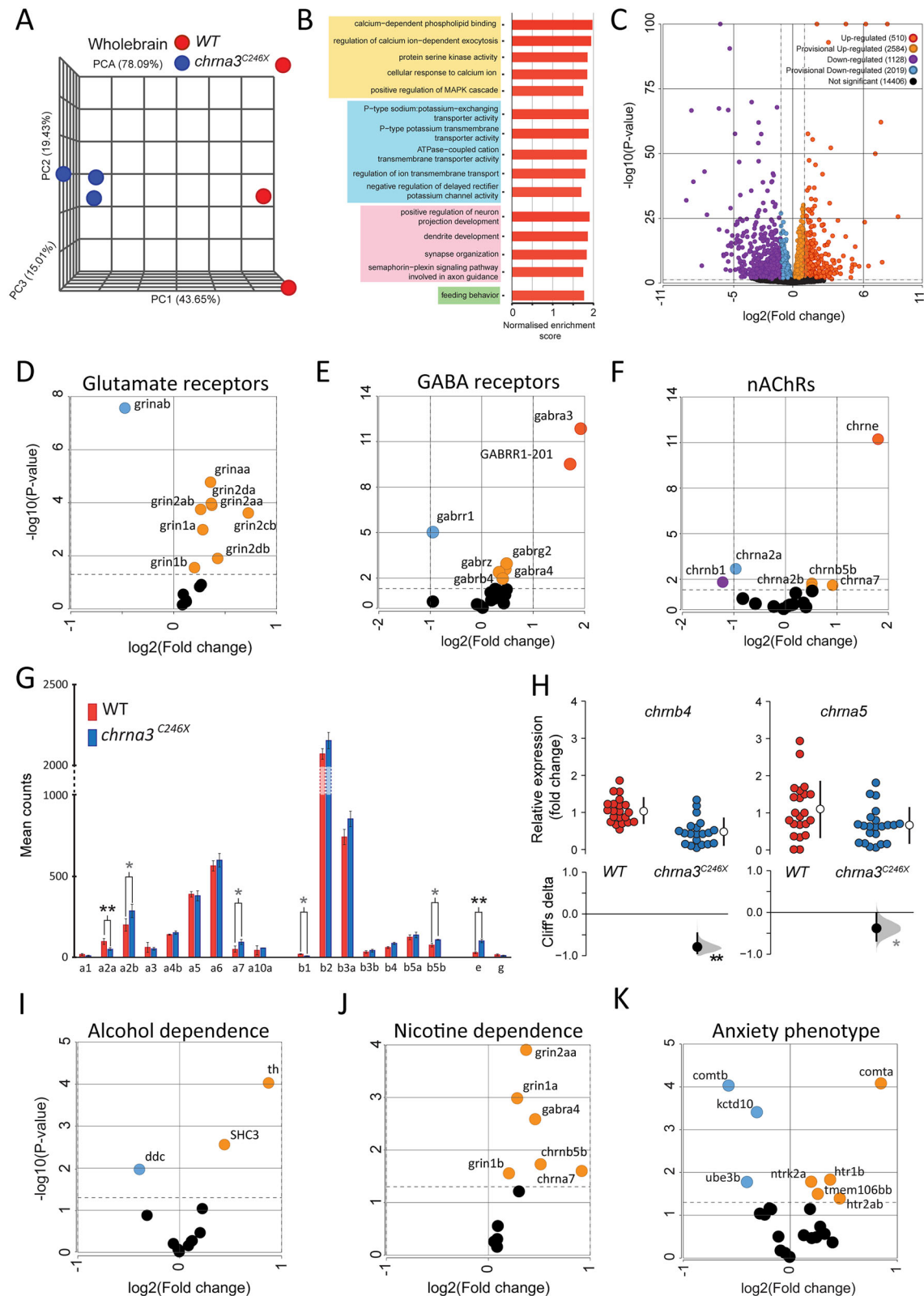


Figure 5. *chrna3*^{C246X} mutants exhibit transcriptional changes in dependence, neuronal development, and ion channel process-associated genes. RNA-seq comparison of *chrna3*^{C246X} (*n* = 3) versus *chrna3*^{C246X} (*n* = 13) 12-week-old zebrafish whole-brain tissues. **A**, PCA clustering of samples by genotype. **B**, All significantly enriched GO terms between genotypes. Positive enrichment values indicate enrichment in the *chrna3*^{C246X} mutant zebrafish. Significance thresholds are defined at *p* ≤ 0.05 and FDR < 0.2. Yellow, secondary messenger associated; blue, ion channel associated; pink, neuronal differentiation and development; green, behaviors. **C**, The volcano plot of gene expression changes between genotypes. Up-/downregulated genes, *p* < 0.05, and log₂ fold change, > ± 2. Provisional up-/downregulated genes, *p* < 0.05, and log₂ fold change, < ± 2. Not significant genes, *p* > 0.05. The color key applies to **D–F** and **I–K**. Positive fold change values indicate upregulation in the *chrna3*^{C246X} mutant zebrafish. **D–F**, Subsets of the volcano plot (**C**), highlighting (**D**) glutamate, (**E**) GABA, and (**F**) nicotinic acetylcholine (nAChR) receptor gene families. **G**, Mean counts (± standard deviation) of nAChRs (gene prefix "chrn"). Stars indicate significant difference between genotypes, ***p* < 0.01; **p* < 0.05. **H**, Fold change in WT versus *chrna3*^{C246X} of *chrna5* (*n* = 21/20) and *chrnb4* (*n* = 22/22) by qRT-PCR. Stars indicate significant difference between genotypes, ***p* < 0.01; **p* < 0.05. **I–K**, Subsets of a volcano plot (**C**), highlighting genes associated with (**I**) nicotine or (**J**) alcohol dependence (Hu et al., 2018) and (**K**) anxiety phenotypes (Howe et al., 2016; Koskinen and Hovatta, 2023). The full gene lists can be found in Table 4.

Discussion

In this study, a refinement of SAZA revealed an acute biphasic response to alcohol, in juvenile zebrafish, with preference transitioning to avoidance when exposure increased to ~0.7% alcohol (Fig. 2). This biphasic response to alcohol aligns zebrafish with other animals, suggesting that hormesis to alcohol is a fundamental property across vertebrates (Figs. 1, 2; Calabrese and Baldwin, 2003; Calabrese, 2008; Sterling et al., 2015). The closed-loop design of SAZA, similar to a two-bottle choice assay used for rodents, facilitated the analysis of uninhibited choice and isolated *chrna3*'s role in modulating this biphasic response.

Isolating contributions of individual genetic factors to a phenomenon such as substance dependence is challenging. It is further compounded for multisubunit receptor genes that cluster at a genomic locus, such as the *CHRNA5-A3-B4* gene cluster. This genomic locus has been consistently associated with several SUD GWASes (Chen et al., 2009; Maes et al., 2011; Haller et al., 2014; Wen et al., 2016). Although male transgenic mice overexpressing *Chrna5-a3-b4* consume less alcohol than WT mice (Gallego et al., 2012), a causal relationship can't be ascribed to *Chrna3* alone from this study (Kamens et al., 2023). *Chrna3* knock-out mice die prematurely as neonates (Xu et al., 1999), and the conditional knock-out (Elayouby et al., 2021) only examined nicotine response leaving the role of $\alpha 3$ in alcohol stimulation unclear (Kamens et al., 2009). Zebrafish *chrna3* mutants, however, survive to adulthood, offering a tractable system to investigate gene-specific effects here. *chrna3* mutants displayed a blunted aversive switch in the biphasic response while maintaining early attraction (Figs. 2, 3). WT fish exhibited aversive behavior to 0.7% or higher alcohol stimulus, which only became observable in mutants at higher concentrations. This aligns with rodent studies, where a broad-spectrum nicotinic receptor antagonist 18-MC in the MHB-IPN circuit reduced transmission and increased nicotine or alcohol self-administration, respectively (Glick et al., 2011; Miller et al., 2019). As this circuit coexpresses all the three nAChR subunits $\alpha 3$, $\alpha 5$, and $\beta 4$, they are thus proposed to mediate aversion to psychoactive substances, including alcohol (Marks et al., 1992; Xu et al., 2006; Frahm et al., 2011; Fowler and Kenny, 2012; Mathuru, 2018; Elayouby et al., 2021). However, as shown in the gene expression profile here (Fig. 1), and in a recent report (Hua et al., 2025), *chrna3* is expressed widely in the central and peripheral nervous system. Consequently, altered alcohol tolerance reported here may stem from a combined effect across multiple circuits and systems. Further to this limitation, although most physiological features and brain regions described in adult zebrafish are present in juveniles by 25 dpf, late-stage developmental changes may influence adult phenotypes (Schmidt et al., 2013; Singleman and Holtzman, 2014; Raj et al., 2020), warranting further studies.

A shifted tolerance window exhibited by *chrna3* mutants (Fig. 3C) suggests a threshold-based engagement of different brain circuits, similar to the proposed mammalian twin threshold model (Grasing, 2016). This model posits that cholinergic signaling, particularly in the VTA and nucleus accumbens, modulates drug reinforcement behaviors. In this model, a lower threshold increases positive reward association probability, stimulating behavioral repetition, while a second, higher threshold does the opposite as different circuits come into play (Grasing, 2016). This may also be applicable to other cholinergic brain regions, such as the MHB. Additionally, higher agonist requirements to activate $\alpha 5$, $\alpha 3$, and $\beta 4$ containing nAChRs (Coe et al., 2005) could elevate the cholinergic threshold to engage this aversive

circuit, providing a potential explanation for why *chrna3* mutation predominantly affected the avoidance response.

Beyond preference, *chrna3* mutation also blunted the psychoactive effects of alcohol during immersion. Immersion is a rapid, noninvasive route for administration of water-soluble pharmaceuticals, uniquely available for aquatic animals. Alcohol immersion time correlates with BAC until equilibrium is reached and is a key measure affecting physiological responses (Lockwood et al., 2004). As in humans, BAC >0.05% or the equivalence of ~2–3 Australian “standard drinks” results in significant behavioral impairment in zebrafish, which can be reached within 1–2 min of immersion in 1.5% alcohol. This manifests in humans as slower reaction times (Grant et al., 2000) and zebrafish as loss of swimming coordination or impaired locomotion (Tran and Gerlai, 2013; Tsang et al., 2019). However, lower concentrations are stimulatory, activating dopaminergic reward pathways and increasing locomotion while reducing anxiety-like behaviors and aggression (Echevarria et al., 2010; Araujo-Silva et al., 2020).

Shoaling behavior, which becomes prominent by 14 dpf (Miller and Gerlai, 2007), can effectively model these contrasting effects (Oscar-Berman and Marinković, 2007; Kurta and Palestis, 2010; Miller et al., 2013; Rueger and King, 2013) at SAZA-relevant concentrations. In WTs, both 0.5 and 1% alcohol immersion gradually reduced shoal cohesion but exhibited differing effects on shoal kinesis (Fig. 4). One interpretation of the coupled reduction in cohesion and kinesis in 1% alcohol is as a sedative effect, evident also in the video recordings (Movie 2). Meanwhile, reduction of cohesion alone could indicate an anxiolytic effect that dominated in a narrow alcohol concentration range (0.5%). These concentrations approximately correlated with the attractive and aversive doses during SAZA, suggesting a potential link. However, the *chrna3*^{C246X} mutants displayed no reduction in shoal cohesion at 0.5% (Fig. 4H). Whether this reduction in anxiolysis phenotype in the mutants was directly due to reduced sensitivity to alcohol or independent of it needs further investigation. Irrespective, it is evident that the *chrna3* mutation blunted drug-induced behavioral effects, including the reduced aversion in SAZA.

Alcohol-induced kinesis also differed between genotypes, with *chrna3*^{C246X} mutants exhibiting stronger alcohol-induced locomotion in SAZA and shoaling assays (Fig. 4, Fig. S5). This may be correlated to the reduced expression of *chrna3* in the caudal and rostral regions of the mutant Fmn, also termed the octavolateral efferent nucleus (Fig. 1, Fig. S1), a hindbrain region connected to the lateral line. Cholinergic signaling here is hypothesized to act as a “brake” on locomotor aut sensory stimulation generated during swimming (Pichler and Lagnado, 2020; Manuel et al., 2021; Odstrcil et al., 2022). While typically this function is assigned to $\alpha 9$ receptors, if $\alpha 3$ plays a role, as suggested recently (Rima et al., 2020), it may account for the heightened alcohol-induced locomotor response in *chrna3*^{C246X} mutants. However, *chrna3*'s influence on locomotor control requires further investigation, as rodent studies offer inconsistent evidence (Kamens et al., 2009).

Whole-body mutation in our *chrna3* mutant precluded implicating specific brain regions like the dorsal habenula (MHB in mammals), yet it does inform on potential consequences of *CHRNA3* variants that reduce $\alpha 3$ function in humans. Whole-brain RNA-seq identified broad transcriptional profile change in the mutants. Small but consistent upregulation of nicotine dependence-associated genes was indicative of a bidirectional relationship (Fig. 5J; Hu et al., 2018), aligning with GWAS studies linking *CHRNA3* polymorphisms to increased

nicotine dependence (Chen et al., 2009; Maes et al., 2011). Downregulation of *chrna5* and *chrnb4* (Fig. 5H), also associated with substance dependence (Fowler et al., 2011; Frahm et al., 2011) and coexpressed with *chrna3* in other vertebrates, potentially contributed to the mutant's alcohol tolerance. Generation and characterization of an independent *chrna3*, *chrna5*, and *chrnb4* mutant lines along with their combinations (currently in progress) will be invaluable to dissociate these factors further.

Beyond nAChRs, alcohol impacts both glutamate and GABA receptors, modulating excitatory and inhibitory tones, respectively (Davies, 2003; Möykkynen and Korpi, 2012; Jones, 2019). Consequently, changes in expression of these receptor genes (Fig. 5D,E) in mutants was also interesting to observe, as many homologs also feature in nicotine dependence and anxiety phenotypes in humans (Fig. 5J,K). At a functional level, neuronal receptor-related GO terms like synaptic signaling, ion channel function, and cation homeostasis were significantly enriched (Fig. 5B). However, neural development terms including neuron projection, dendrite development, and axon guidance were also enriched despite the use of adult brains (Fig. 5B), perhaps indicative of compensatory plasticity. Finally, enrichment of feeding behavior-associated genes was notable (Fig. 5B), as cholinergic signaling and CHRNA5-CHRNA3-CHRNA4 cluster SNPs have been tentatively linked by GWAS studies to smoking/obesity interactions (Varga et al., 2013; Herman et al., 2016; Ortiz-Guzman et al., 2024). This transcriptomic change warrants further examination to determine the simultaneous influence of *chrna3* on substance dependence and other comorbid disorders.

Overall, these findings suggest the type of brain-wide compensatory changes that may occur corresponding to loss of *chrna3* function, eventually resulting in heightened alcohol tolerance and potential predisposition to multisubstance use and comorbid disorders.

Understanding alcohol's biphasic properties is also imperative to address AUD. Our study thus fills a gap, providing novel evidence on the role of *chrna3* in modulating alcohol preference, and tethering it to a specific facet of the dose–response curve. Given the paucity of animal model data, despite GWAS associations (Haller et al., 2014), the evidence of isolated gene function from a whole-body mutation in zebrafish presented here represents a valuable step forward in the context of phenotypes leading to alcohol dependence development in humans harboring CHRNA3 variants.

Data Availability

The datasets used to generate the figures are included within the article and its additional file(s). Additional data are available from the corresponding author upon request.

References

- Amos CI, Spitz MR, Cinciripini P (2010) Chipping away at the genetics of smoking behavior. *Nat Genet* 42:366–368.
- Araujo-Silva H, Leite-Ferreira ME, Luchiarini AC (2020) Behavioral screening of alcohol effects and individual differences in zebrafish (*Danio rerio*). *Alcohol Alcohol* 55:591–597.
- Aslam A, Kwo PY (2023) Epidemiology and disease burden of alcohol associated liver disease. *J Clin Exp Hepatol* 13:88–102.
- Banerjee TD, Raine J, Mathuru AS, Chen KH, Monteiro A (2024) Spatial mRNA profiling using rapid amplified multiplexed-FISH (RAM-FISH). *bioRxiv*:2024.12.06.627193. Available at: <https://www.biorxiv.org/content/10.1101/2024.12.06.627193v1.abstract> [Accessed May 5, 2025].
- Bassett AR, Tibbit C, Ponting CP, Liu J-L (2013) Highly efficient targeted mutagenesis of *Drosophila* with the CRISPR/Cas9 system. *Cell Rep* 4:220–228.
- Bierut LJ (2011) Genetic vulnerability and susceptibility to substance dependence. *Neuron* 69:618–627.
- Bogdan R, Hatoum AS, Johnson EC, Agrawal A (2023) The Genetically Informed Neurobiology of Addiction (GINA) model. *Nat Rev Neurosci* 24:40–57.
- Bossé GD, Peterson RT (2017) Development of an opioid self-administration assay to study drug seeking in zebrafish. *Behav Brain Res* 335:158–166.
- Calabrese EJ (2008) U-shaped dose response in behavioral pharmacology: historical foundations. *Crit Rev Toxicol* 38:591–598.
- Calabrese EJ, Baldwin LA (2003) Ethanol and hormesis. *Crit Rev Toxicol* 33:407–424.
- Chen X, Chen J, Williamson VS, An S-S, Hettema JM, Aggen SH, Neale MC, Kendler KS (2009) Variants in nicotinic acetylcholine receptors alpha5 and alpha3 increase risks to nicotine dependence. *Am J Med Genet B Neuropsychiatr Genet* 150B:926–933.
- Choi HMT, Schwarzkopf M, Fornace ME, Acharya A, Artavanis G, Stegmaier J, Cunha A, Pierce NA (2018) Third-generation *in situ* hybridization chain reaction: multiplexed, quantitative, sensitive, versatile, robust. *Development* 145:dev165753.
- Chvilicek MM, Titos I, Rothenfluh A (2020) The neurotransmitters involved in alcohol-induced behaviors. *Front Behav Neurosci* 14:607700.
- Cliff N (1993) Dominance statistics: ordinal analyses to answer ordinal questions. *Psychol Bull* 114:494–509.
- Coe JW, et al. (2005) Varenicline: an alpha4beta2 nicotinic receptor partial agonist for smoking cessation. *J Med Chem* 48:3474–3477.
- Cox TC (2015) Utility and limitations of animal models for the functional validation of human sequence variants. *Mol Genet Genomic Med* 3:375–382.
- Cully M (2019) Zebrafish earn their drug discovery stripes. *Nat Rev Drug Discov* 18:811–813.
- Davies M (2003) The role of GABAA receptors in mediating the effects of alcohol in the central nervous system. *J Psychiatry Neurosci* 28:263–274.
- de Magalhães JP (2015) The big, the bad and the ugly: extreme animals as inspiration for biomedical research. *EMBO Rep* 16:771–776.
- Echevarria DJ, Hammack CM, Jouandot DJ, Toms CN (2010) Does acute alcohol exposure modulate aggressive behaviors in the zebrafish (*Danio rerio*), or is the bark worse than the bite? *Int J Comp Psychol* 23:62–69.
- Echevarria DJ, Toms CN, Jouandot DJ (2011) Alcohol-induced behavior change in zebrafish models. *Rev Neurosci* 22:85–93.
- Egervari G, Siciliano CA, Whiteley EL, Ron D (2021) Alcohol and the brain: from genes to circuits. *Trends Neurosci* 44:1004–1015.
- Elayoubi KS, Ishikawa M, Dukes AJ, Smith ACW, Lu Q, Fowler CD, Kenny PJ (2021) A3* Nicotinic acetylcholine receptors in the habenula-interpeduncular nucleus circuit regulate nicotine intake. *J Neurosci* 41:1779–1787.
- Fowler CD, Kenny PJ (2012) Habenular signaling in nicotine reinforcement. *Neuropsychopharmacology* 37:306–307.
- Fowler CD, Lu Q, Johnson PM, Marks MJ, Kenny PJ (2011) Habenular α5 nicotinic receptor subunit signalling controls nicotine intake. *Nature* 471:597–601.
- Frahm S, et al. (2011) Aversion to nicotine is regulated by the balanced activity of β4 and α5 nicotinic receptor subunits in the medial habenula. *Neuron* 70:522–535.
- Gallego X, Ruiz-Medina J, Valverde O, Molas S, Robles N, Sabrià J, Crabbe JC, Dierssen M (2012) Transgenic over expression of nicotinic receptor alpha 5, alpha 3, and beta 4 subunit genes reduces ethanol intake in mice. *Alcohol* 46:205–215.
- Gebauer DL, Pagnussat N, Piatto AL, Schaefer IC, Bonan CD, Lara DR (2011) Effects of anxiolytics in zebrafish: similarities and differences between benzodiazepines, buspirone and ethanol. *Pharmacol Biochem Behav* 99:480–486.
- Gerlai R (2020) Evolutionary conservation, translational relevance and cognitive function: the future of zebrafish in behavioral neuroscience. *Neurosci Biobehav Rev* 116:426–435.
- Glick SD, Sell EM, McCallum SE, Maisonneuve IM (2011) Brain regions mediating α3β4 nicotinic antagonist effects of 18-MC on nicotine self-administration. *Eur J Pharmacol* 669:71–75.
- Goodman WM, Spruill SE, Komaroff E (2019) A proposed hybrid effect size plus *p*-value criterion: empirical evidence supporting its use. *Am Stat* 73:168–185.
- Gotti C, Zoli M, Clementi F (2006) Brain nicotinic acetylcholine receptors: native subtypes and their relevance. *Trends Pharmacol Sci* 27:482–491.
- Grant SA, Millar K, Kenny GN (2000) Blood alcohol concentration and psychomotor effects. *Br J Anaesth* 85:401–406.

- Grasing K (2016) A threshold model for opposing actions of acetylcholine on reward behavior: molecular mechanisms and implications for treatment of substance abuse disorders. *Behav Brain Res* 312:148–162.
- Haller G, et al. (2014) Rare missense variants in *CHRNA3* and *CHRNA5* are associated with risk of alcohol and cocaine dependence. *Hum Mol Genet* 23:810–819.
- Hellemans J, Mortier G, De Paepe A, Speleman F, Vandesompele J (2007) qBase relative quantification framework and software for management and automated analysis of real-time quantitative PCR data. *Genome Biol* 8:R19.
- Herman AM, et al. (2016) A cholinergic basal forebrain feeding circuit modulates appetite suppression. *Nature* 538:253–256.
- Ho J, Tumkaya T, Aryal S, Choi H, Claridge-Chang A (2019) Moving beyond *p* values: data analysis with estimation graphics. *Nat Methods* 16:565–566.
- Howe AS, et al. (2016) Candidate genes in panic disorder: meta-analyses of 23 common variants in major anxiogenic pathways. *Mol Psychiatry* 21:665–679.
- Hu R, Dai Y, Jia P, Zhao Z (2018) ANCO-GeneDB: annotations and comprehensive analysis of candidate genes for alcohol, nicotine, cocaine and opioid dependence. *Database* 2018:bay121.
- Hua Y, Habicher J, Carl M, Manuel R, Boije H (2025) Novel transgenic zebrafish lines to study the *CHRNA3-B4-A5* gene cluster. *Dev Neurobiol* 85: e22956.
- Hwang WY, Fu Y, Reyon D, Maeder ML, Tsai SQ, Sander JD, Peterson RT, Yeh J-RJ, Joung JK (2013) Efficient genome editing in zebrafish using a CRISPR-Cas system. *Nat Biotechnol* 31:227–229.
- Jones AW (2019) Alcohol, its absorption, distribution, metabolism, and excretion in the body and pharmacokinetic calculations. *WIREs Forensic Sci* 1:e1340.
- Kamens HM, McKinnon CS, Li N, Helms ML, Belknap JK, Phillips TJ (2009) The alpha 3 subunit gene of the nicotinic acetylcholine receptor is a candidate gene for ethanol stimulation. *Genes Brain Behav* 8:600–609.
- Kamens HM, Flarend G, Horton WJ (2023) The role of nicotinic receptors in alcohol consumption. *Pharmacol Res* 190:106705.
- Kibat C (2020) Functional characterization of nicotinic acetylcholine receptor subunit ALPHA 3 in regulation of behaviour.
- Kily LJM, Cowe YCM, Hussain O, Patel S, McElwaine S, Cotter FE, Brennan CH (2008) Gene expression changes in a zebrafish model of drug dependency suggest conservation of neuro-adaptation pathways. *J Exp Biol* 211: 1623–1634.
- Koob GF, Vendruscolo L (2023) Theoretical frameworks and mechanistic aspects of alcohol addiction: alcohol addiction as a reward deficit/stress surfeit disorder. *Curr Top Behav Neurosci* 71:1–61.
- Koskinen M-K, Hovatta I (2023) Genetic insights into the neurobiology of anxiety. *Trends Neurosci* 46:318–331.
- Krishnan S, Mathuru AS, Kibat C, Rahman M, Lupton CE, Stewart J, Claridge-Chang A, Yen S-C, Jesuthasan S (2014) The right dorsal habenula limits attraction to an odor in zebrafish. *Curr Biol* 24:1167–1175.
- Kunst M, Laurell E, Mokayes N, Kramer A, Kubo F, Fernandes AM, Förster D, Dal Maschio M, Baier H (2019) A cellular-resolution atlas of the larval zebrafish brain. *Neuron* 103:21–38.e5.
- Kurta A, Palestis BG (2010) Effects of ethanol on the shoaling behavior of zebrafish (*Danio rerio*). *Dose Response* 8(4):527–533. Available at: <https://journals.sagepub.com/doi/10.2203/dose-response.10-008.Palestis> [Accessed Oct. 17, 2023].
- Lockwood B, Bjerke S, Kobayashi K, Guo S (2004) Acute effects of alcohol on larval zebrafish: a genetic system for large-scale screening. *Pharmacol Biochem Behav* 77:647–654.
- Maes HH, Neale MC, Chen X, Chen J, Prescott CA, Kendler KS (2011) A twin association study of nicotine dependence with markers in the *CHRNA3* and *CHRNA5* genes. *Behav Genet* 41:680–690.
- Manuel R, Iglesias Gonzalez AB, Habicher J, Koning HK, Boije H (2021) Characterization of individual projections reveal that neuromasts of the zebrafish lateral line are innervated by multiple inhibitory efferent cells. *Front Neuroanat* 15:666109.
- Marks MJ, Pauly JR, Gross SD, Deneris ES, Hermans-Borgmeyer I, Heinemann SF, Collins AC (1992) Nicotine binding and nicotinic receptor subunit RNA after chronic nicotine treatment. *J Neurosci* 12:2765–2784.
- Mathur P, Berberoglu MA, Guo S (2011) Preference for ethanol in zebrafish following a single exposure. *Behav Brain Res* 217:128–133.
- Mathuru AS (2018) A little rein on addiction. *Semin Cell Dev Biol* 78:120–129.
- Miller CN, Ruggery C, Kamens HM (2019) The $\alpha 3\beta 4$ nicotinic acetylcholine receptor antagonist 18-methoxycoronaridine decreases binge-like ethanol consumption in adult C57BL/6J mice. *Alcohol* 79:1–6.
- Miller N, Gerlai R (2007) Quantification of shoaling behaviour in zebrafish (*Danio rerio*). *Behav Brain Res* 184:157–166.
- Miller N, Greene K, Dydzinski A, Gerlai R (2013) Effects of nicotine and alcohol on zebrafish (*Danio rerio*) shoaling. *Behav Brain Res* 240:192–196.
- Moeller G, Terplan M (2020) *Substance use disorders*. USA: Oxford University Press.
- Möykkynen T, Korpi ER (2012) Acute effects of ethanol on glutamate receptors. *Basic Clin Pharmacol Toxicol* 111:4–13.
- Nathan FM, Kibat C, Goel T, Stewart J, Claridge-Chang A, Mathuru AS (2022) Contingent stimulus delivery assay for zebrafish reveals a role for *CCSER1* in alcohol preference. *Addict Biol* 27:e13126.
- Odstrcil I, et al. (2022) Functional and ultrastructural analysis of reafferent mechanosensation in larval zebrafish. *Curr Biol* 32:176–189.e5.
- Ortiz-Guzman J, et al. (2024) Cholinergic basal forebrain connectivity to the basolateral amygdala modulates food intake. *eNeuro* 11:369.
- Oscar-Berman M, Marinković K (2007) Alcohol: effects on neurobehavioral functions and the brain. *Neuropsychol Rev* 17:239–257.
- Pichler P, Lagnado L (2020) Motor behavior selectively inhibits hair cells activated by forward motion in the lateral line of zebrafish. *Curr Biol* 30:150–157.e3.
- Raj B, et al. (2020) Emergence of neuronal diversity during vertebrate brain development. *Neuron* 108:1058–1074.e6.
- Rehm J, Shield KD (2019a) Global burden of disease and the impact of mental and addictive disorders. *Curr Psychiatry Rep* 21:10.
- Rehm J, Shield KD (2019b) Global burden of alcohol use disorders and alcohol liver disease. *Biomedicine* 7:99.
- Rima M, Lattouf Y, Abi Younes M, Bullier E, Legendre P, Mangin J-M, Hong E (2020) Dynamic regulation of the cholinergic system in the spinal central nervous system. *Sci Rep* 10:15338.
- Rueger SY, King AC (2013) Validation of the brief biphasic alcohol effects scale (B-BAES). *Alcohol Clin Exp Res* 37:470–476.
- Schmidt R, Strähle U, Scholpp S (2013) Neurogenesis in zebrafish - from embryo to adult. *Neural Dev* 8:3.
- Schneider H, Pearson's A, Harris D, Tucker A, Gardner K, Chinyanya K (2023) Identification of nicotine-seeking and avoiding larval zebrafish using a new three-choice behavioral assay. *Front Mol Neurosci* 16:112927.
- Singleman C, Holtzman NG (2014) Growth and maturation in the zebrafish, *Danio rerio*: a staging tool for teaching and research. *Zebrafish* 11:396–406.
- Smith MA (2020) Nonhuman animal models of substance use disorders: translational value and utility to basic science. *Drug Alcohol Depend* 206:107733.
- Sterling ME, Karatayev O, Chang G-Q, Algava DB, Leibowitz SF (2015) Model of voluntary ethanol intake in zebrafish: effect on behavior and hypothalamic orexigenic peptides. *Behav Brain Res* 278:29–39.
- Thorgeirsson TE, et al. (2008) A variant associated with nicotine dependence, lung cancer and peripheral arterial disease. *Nature* 452:638–642.
- Tran S, Gerlai R (2013) Time-course of behavioural changes induced by ethanol in zebrafish (*Danio rerio*). *Behav Brain Res* 252:204–213.
- Tsang B, Ansari R, Gerlai R (2019) Dose dependent behavioral effects of acute alcohol administration in zebrafish fry. *Pharmacol Biochem Behav* 179: 124–133.
- Varga TV, Hallmans G, Hu FB, Renström F, Franks PW (2013) Smoking status, snus use, and variation at the *CHRNA5-CHRNA3-CHRNA4* locus in relation to obesity: the GLACIER study. *Am J Epidemiol* 178:31–37.
- Wada E, Wada K, Boulter J, Deneris E, Heinemann S, Patrick J, Swanson LW (1989) Distribution of alpha 2, alpha 3, alpha 4, and beta 2 neuronal nicotinic receptor subunit mRNAs in the central nervous system: a hybridization histochemical study in the rat. *J Comp Neurol* 284:314–335.
- Wen L, Jiang K, Yuan W, Cui W, Li MD (2016) Contribution of variants in *CHRNA5/A3/B4* gene cluster on chromosome 15 to tobacco smoking: from genetic association to mechanism. *Mol Neurobiol* 53:472–484.
- Witkiewitz K, Litten RZ, Leggio L (2019) Advances in the science and treatment of alcohol use disorder. *Sci Adv* 5:eaax4043.

- Xu W, Gelber S, Orr-Urtreger A, Armstrong D, Lewis RA, Ou CN, Patrick J, Role L, De Biasi M, Beaudet AL (1999) Megacystis, mydriasis, and ion channel defect in mice lacking the alpha3 neuronal nicotinic acetylcholine receptor. *Proc Natl Acad Sci U S A* 96:5746–5751.
- Xu X, Scott MM, Deneris ES (2006) Shared long-range regulatory elements coordinate expression of a gene cluster encoding nicotinic receptor heteromeric subtypes. *Mol Cell Biol* 26:5636–5649.
- Yang W, Singla R, Maheshwari O, Fontaine CJ, Gil-Mohapel J (2022) Alcohol use disorder: neurobiology and therapeutics. *Biomedicines* 10:1192.
- Zhou X, Zheng W, Li Y, Pearce R, Zhang C, Bell EW, Zhang G, Zhang Y (2022) I-TASSER-MTD: a deep-learning-based platform for multi-domain protein structure and function prediction. *Nat Protoc* 17:2326–2353.